

Review

Factors Influencing the Impact of Anaerobic Digestates on Soil Properties

Péter Ragályi , Orsolya Szécsy *, Nikolett Uzinger, Marianna Magyar , Anita Szabó  and Márk Rékási

Institute for Soil Sciences, Centre for Agricultural Research, HUN-REN, H-1116 Budapest, Hungary; uzinger.nikolett@atk.hun-ren.hu (N.U.); magyar.marianna@atk.hun-ren.hu (M.M.); szabo.anita@atk.hun-ren.hu (A.S.); rekasi.mark@atk.hun-ren.hu (M.R.)

* Correspondence: ragalyi.peter@atk.hun-ren.hu (P.R.); szecsy.orsolya@atk.hun-ren.hu (O.S.)

Abstract

Green energy is expected to play an increasingly important role in the energy sector, so the volume of biogas production and the formation of anaerobic digestates is likely to increase in the future. A wide range of biodegradable organic materials are used in anaerobic digesters to produce biogas. This review focuses on the properties of anaerobic digestates and their effects on physical, chemical and biological soil parameters discussing the benefits, limitations and potential risks. Due to the variety of technologies and raw materials used, anaerobic digestates have diverse properties. Therefore, their impact on specific soil parameters, such as bulk density, aggregate stability, pH, electrical conductivity (EC), soil organic matter (SOM) or microbial activity can vary in magnitude and direction. These effects are also influenced by the variety of soils. Although digestates usually have a significant macro- and micronutrient content, their potentially toxic components or high salt content may limit their use. Despite the limitations, the application of anaerobic digestates generally has more advantages than disadvantages. The use of good-quality anaerobic digestates can improve the physical and chemical properties of the soil, increase soil nutrient and SOM content, as well as soil microbial activity.

Keywords: biogas slurry; liquid fermentate; soil amendment; organic fertilizer; fermentation residue; circular economy; soil characteristics; soil properties; pollutants



Received: 20 May 2025

Revised: 3 July 2025

Accepted: 11 July 2025

Published: 14 July 2025

Citation: Ragályi, P.; Szécsy, O.; Uzinger, N.; Magyar, M.; Szabó, A.; Rékási, M. Factors Influencing the Impact of Anaerobic Digestates on Soil Properties. *Soil Syst.* **2025**, *9*, 78. <https://doi.org/10.3390/soilsystems9030078>

Copyright: © 2025 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (<https://creativecommons.org/licenses/by/4.0/>).

1. Introduction

It is estimated that in 2017, approximately 20 billion tons of waste were generated worldwide in one year, which is expected to increase to 46 billion tons by 2050 [1]. The amount of municipal solid waste generated in one year was 7–9 billion tons in 2023, which is also expected to increase in the future [2]. On average, 511 kg of solid waste was generated per capita in European countries in 2023 [3], of which 104.5 kg was biodegradable [4]. Nearly one-third of food intended for human consumption ends up as waste [5]. Landfilling of waste with usable components and energy content should be minimized, as it burdens the environment and can act as a source of pollution. Biological waste is valuable in terms of both its composition and energy content, as it can be treated biologically: it can be composted, anaerobically digested, but it can also be used for direct heat generation via incineration [6]. During composting and anaerobic digestion, organic waste is transformed by microbes into simpler, lower-energy and more stable organic materials [7]. Composting, vermicomposting, anaerobic digestion and pyrolysis are the most common treatments

applied to increase the valuable plant nutrients in organic wastes or by-products in order to achieve higher plant biomass production and nutrient recycling [8].

During anaerobic digestion, biogas is produced from organic waste, and the digestion residue can be further used for fertilization or as soil amendment. Almost any feedstock containing biodegradable organic matter (OM) can be used to produce biogas through anaerobic digestion. The raw material and the processing technology have a significant impact on the properties of the resulting digestate [9].

According to a study published in 2022, there are approximately 50 million anaerobic digesters of various capacities in the world. The number of digesters is increasing worldwide, but in Europe this rate exceeds the world average. Based on their capacity, European digesters process 10–20 thousand tons of waste per year [10]. Jurgutis et al. [11] suggest that a supportive regulatory environment will accelerate the ongoing expansion of biogas plants and encourage investment through the implementation of the EU Green Deal.

For the economical operation of biogas plants, it is essential to have suitable, cheap organic material, the transportation of which to the biogas plant does not represent an economic barrier, i.e., the plant must be in adequate proximity to the resources, the distance of which must usually be within 40 km [12]. The other bottleneck is the resulting digestate, which, if produced in too large quantity compared to the possibility of application to arable lands as a fertilizer or soil amendment, may require various post-treatments for the recovery of nutrients [13].

The circular economy replaces the traditional ‘cradle-to-grave’ linear economic concept with a ‘cradle-to-cradle’ concept that aims to maximize the life cycle of products by utilizing by-products and waste [14]. This includes recovering energy from waste generated in agricultural or urban areas and recycling the by-products of this process—the digestates—as fertilizers [15]. Nowadays, the goal of optimizing anaerobic digestion is primarily to increase gas yield, which is very important for the feasible expansion of green energy, while less attention is paid to the properties of the resulting digestate [16]. A change of perspective is needed to shift the focus from biogas optimization to integrated biogas—digestate optimization [17].

It is necessary to gain a more thorough understanding of the behavior and effects of the digestates on the environment, since they are expected to be produced in increasing quantities. Only a limited number of reviews have been published on the agricultural use of digestates, including their fertilizer value and effect on crop yield [18], optimizing their treatment [19,20], nutrient recovery technics [13], post-treatments for optimal agricultural and non-agricultural valorization [21,22], their impact on soil organisms and soil life [23,24], food-waste digestates [25], and comprehensive reviews on the application of digestate as fertilizer in the previous decade [26,27].

Soil quality, also known as soil health, refers to the ability of soil to function as a vital living ecosystem. This involves maintaining biological productivity and environmental quality, thereby promoting the health of plants, animals and humans in both natural and agricultural ecosystems [28]. Despite the fact that the application of digestates to agricultural soils is a widespread practice, research and especially reviews on their effects on physical, chemical and biological soil parameters, is currently scarce. It is necessary to summarize and synthesize the effects of digestates on soil parameters based on the latest research results for research, education, and agricultural practice.

This present review aims to summarize, compare and study the impact of different digestates on soil properties and the potential limitations of their use. This will provide an overview of the current scientific literature and help to navigate among the issues that are currently relevant or crucial when applying digestates to soil.

2. Methods

The network analysis map was created based on the results obtained from the Web of Science search terms (TS) (anaerobic digestate* OR biogas digestate*) AND (soil) AND (qualit* OR propert* OR characteristic* OR pollut*) (Figure 1). The most important related terms include anaerobic digestion, quality, co-digestion, soil properties, nitrogen (N), phosphorus (P), availability, organic matter, heavy metals, all of which will be discussed in the present review.

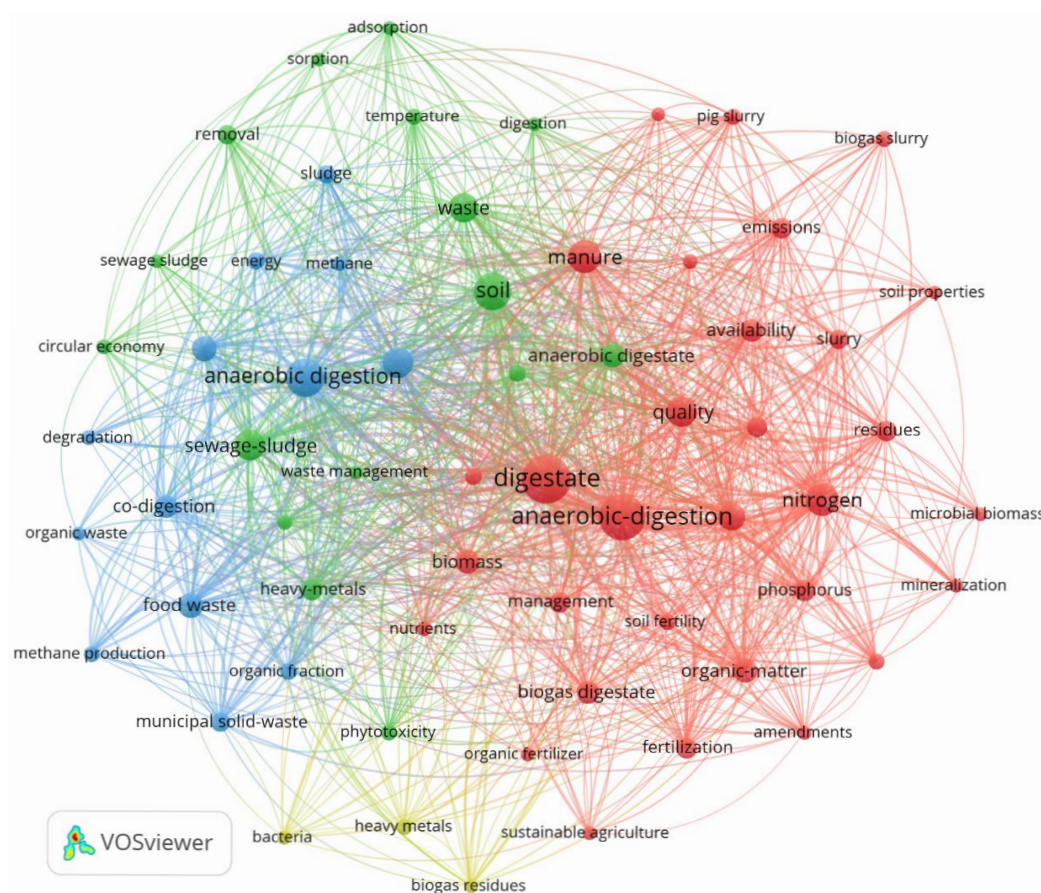


Figure 1. Keyword co-occurrence map indicating the most frequently investigated topics related to anaerobic digestates.

The literature search was conducted using the results of searches in the Web of Science, ScienceDirect, SpringerLink and Google Scholar databases. The sources were selected by the authors of each section or subsection based on their relevance to the topic of the present review. Finally 227 publications published between 2003 and 2025 were used to prepare this review, 70% of the articles were published in the last 10 years (2016–2025), the majority of them were published no earlier than 2019.

3. Effect of Raw Materials on the Quality of Digestates

From a practical point of view, the raw materials for digestion can be divided into various groups, such as animal waste (manure, urine), food waste (from food industry and restaurants), municipal solid waste, municipal sewage sludge and green waste [10].

3.1. Digestates from Animal Waste

Animal manure is one of the most widely used raw materials for biogas production. It contains a large amount of nutrients that are available for plants [29]. Although it is particularly suitable for anaerobic digestion, since it contains several types of microbes, nutrients,

and has a neutral pH, it produces a lower gas yield compared to other raw materials as it is already pre-digested by animals [30]. During anaerobic hydrolysis, approximately 70% of the total N content of the animal manure feedstock is mineralized to form ammonium ions (NH_4^+), or free ammonia (NH_3). Such animal manure-derived digestates typically contain 0.8–5.0 g L⁻¹ ammonium nitrogen ($\text{NH}_4\text{-N}$). Phosphorus (P) comes partly from the feed and partly from inorganic additives, and its concentration can reach 1.2 g L⁻¹ in the digestate. Dissolved P can be detected mostly in the form of orthophosphate (PO_4^{3-}), and its presence is strongly pH-dependent. Approximately 90% of the P content of the digestate forms a precipitate with metals, while the dissolved fraction accounts for only around 10% [31].

The properties of digestates from animal waste may vary depending on the origin of the raw materials (pig, cattle, poultry, etc.). The composition of the digestate also depends on the animal housing technologies, feeding strategies, and manure management [32]. For example, poultry and cattle manure contain a higher proportion of lignified components, which results in slower biodegradability than in the case of pig manure [29,33]. Chicken manure has a higher $\text{NH}_4\text{-N}$ and total N content than cattle or pig manure, and this is also true for the digestate made from it [11]. Digestate produced from pig manure contains approximately 80% of its total N content in the form of NH_4^+ [34]. During anaerobic digestion, organic N is mineralized and NH_4^+ is produced, which can be directly absorbed by crops [26]. The feeding strategy of cattle also has an influence on the digestate. If the animals are fed supplementary corn and soy as well as grazed on grass, the resulting digestate will be richer in nutrients [35].

Pig and cattle manure contain significant amounts of pathogens, most of which are eliminated during the process of anaerobic digestion, but the number of Fecal coliform bacteria may still remain high [34]. According to WHO, such materials can only be used with restrictions. The digestate can be improved by adding CaO after the anaerobic digestion process, and long-term storage also reduces the number of coliform bacteria [36]. In the case of digestates from animal manure and sewage sludge, the presence of pharmaceutical active ingredients and antibiotics can also cause problems for both aquatic and terrestrial ecosystems [37]. Since the various pretreatments are unable to completely remove antibiotics, further research is needed to resolve this issue [38].

3.2. Digestates from Food Waste

The utilization of food-waste anaerobic digestates is of fundamental importance for the realization of a circular bioeconomy [25]. Food waste is an excellent raw material for anaerobic digestion due to its high water content and biodegradability, as well as its excellent methane (CH_4) production potential [39]. The high amount of organic components, carbohydrates, fats and proteins make food waste digestates a valuable fertilizer [40]. Almost 99% of the total N in digestates from food waste may be in the form of NH_4^+ [41]. According to Ren et al. [42], approximately 0.2–0.47 tons of digestate can be produced from 1 ton of food waste.

The properties of food-based digestates greatly depend on the raw material and can vary significantly. Food waste containing a large amount of N-rich material (such as meat) results in digestates with higher N content and are capable of yielding more CH_4 than digestates from animal manure [11]. Food waste containing a higher proportion of plant-based foods results in a digestate with a lower N content [43]. The heavy metal content of food-based digestates is lower than that based on sewage sludge [44], but O'Connor et al. [45] drew attention to the fact that plastics, particulate contaminants, heavy metals and microbial contaminants may significantly limit the use of food waste as a soil amendment. Furthermore, such digestates often have high concentrations of sodium (Na), chlorine

(Cl), P and potassium (K) ions, due to the high salt content of the processed food waste. High Na content may adversely affect the physicochemical properties of the soil and the development of crops [46]. For the longer-term use of digestates made from food waste, regular irrigation or appropriate dilution of the digestate may prevent a detrimental increase in soil salinity and EC values [47].

3.3. Digestates from Municipal Solid Waste

The composition of the municipal solid waste (MSW) can vary significantly across different regions, so it is not possible to generalize its characteristics [48]. The carbon (C), hydrogen (H), moisture and starch content of MSW, and the volatile solids:total solids ratio show only small variability. However, the N, P, sulphur (S), hemicellulose, free sugars, lignin and raw fiber contents may vary greatly. In some countries, food waste and other organic materials account for more than half of all MSW [48].

The digestate formed from the organic fraction of MSW usually has a weak alkaline pH and contains both macro- (N, P, K, Ca, S and Mg) and micronutrients (B, Cl, Mn, Fe, Zn, Cu, Mo and Ni); in some cases, the N, P and K content may be significantly high, but in general the concentration of nutrients is typically lower compared to digestates from manure or sewage sludge [49]. If the organic fraction of the MSW is properly sorted before entering the digester, it may contain negligible amounts of impurities [17].

The high (30–50%) total solids content, high C:N ratio, and non-biodegradable components in the organic fraction of MSW may complicate the anaerobic digestion process. Agricultural use can also be limited by contaminants, but these can be greatly reduced by careful sorting and separation of the raw materials [38]. Physical contaminants are also typical of digestates from MSW. The most typical physical contaminants include plastic, metal, rubber, glass, ceramics, sand and stones, as well as materials with a high cellulose content, such as wood and paper [17].

3.4. Digestates from Sewage Sludge

Wastewater contains a large amount of OM, which makes it suitable for energy production. Primary settling tanks are designed to provide a suitable substrate for biogas production even in the wastewater treatment phase, so the proportion of easily biodegradable C is also controlled [50]. Some estimates suggest that the amount of P in wastewater may reach 16% of the P annually mined. In wastewater treatment plants, the uncontrolled precipitation of P is a common problem, causing a reduction in the operating volume of anaerobic digesters and higher operating costs due to the need for acid flushing and increased downtime [31].

Digested sewage sludge typically contains 50% OM. Its total N content is usually 1–6%, but the processing parameters significantly influence this value. Its content of essential nutrients (N, P, S) and micronutrients makes it suitable for use as a fertilizer. The hazardous components in sewage sludge, such as heavy metals and persistent organic pollutants, cannot be removed by anaerobic digestion, so appropriate treatment may be required to avoid environmental and public health issues [51]. The presence of pathogens, pharmaceuticals, cosmetics, and surfactants can also cause environmental damage [6].

3.5. Digestates from Green Waste

Green wastes are usually difficult to decompose due to their high C:N ratio, so it is recommended to decompose them together with material containing a high level of N to create an optimal C:N ratio of 15–30. The decomposition of woody plant residues may reduce the gas yield, as lignin is less easily decomposed, which reduces efficiency [52].

Plant tissues have a high lignocellulose content, and thus a complex structure making them resistant to decomposition, so the derived digestates also contain semi-decomposed

OM. These residual components, however, can be useful as soil conditioners [38]. Wheat straw pre-treated with potassium hydroxide can be used to make a digestate rich in Ca, Mg and K [53], furthermore corn silage [54] and sugar factory residue [55] are also excellent raw materials. A smaller amount of digestate is produced from plant raw materials with a low fiber content [56], while the digestate formed from an input with higher fiber content also has higher dry matter content [57]. In the case of raw materials with a high lignocellulose content, pre-treatment may be required, in the form of milling, extrusion, steam explosion, microwave, acid or alkaline pretreatment [58]. Digestates produced in biogas plants that use only plant-based ingredients usually do not contain pathogens or heavy metals [38].

3.6. Digestates from Mixed Materials—The Result of Co-Digestion

It is clear from the previous subsections that materials suitable for biogas production may have properties that limit either the anaerobic digestion process or the usability of the digestate. In order to eliminate the disadvantages, promote the decomposition of the raw materials, increase the biogas yield, and improve the quality of the resulting digestate, instead of using a single type of raw material, biogas plants generally co-digest a mixture of raw materials to produce biogas [59].

Care must be taken in the selection of raw materials, because efficiency can be increased with the right combination of materials, while the wrong combination may result in reduced biogas production or even the cessation of the process [30]. In a meta-analysis of 124 articles on livestock manure co-digestion, Zhou et al. [60] found a recommended range for the C:N ratio of 20–27, with a lipid:carbohydrate ratio > 0.13 , and a protein:carbohydrate ratio > 0.26 . The advantages, limitations and current status of co-digestion, and co-digestion with different mixing ratios and feedstocks were summarized by Karki et al. [61], who recommended the co-digestion process especially for its effective environmental and economic sustainability, waste management, and resource utilization. Possible optimal combinations may be a mixture of (i) food waste and green waste, (ii) slurry and green waste, or (iii) municipal solid waste and sewage sludge [62–64].

Koch et al. [65] recommended the use of food waste as a co-substrate in the biogas digesters operated in wastewater treatment plants. The combined use of these two types of materials is beneficial for several reasons. Food waste mixed with raw sewage sludge yields more CH_4 and, as an easily degradable substrate, improves the digestion of sewage sludge, which is much more difficult to decompose. Furthermore, it results in a better supply of macro and trace elements in the digestate, since the unfavorable C:N ratio of raw sludge is improved by the food waste [65].

Animal manures are suitable for biogas production on their own due to their adequate organic components and fairly high buffer capacity. However, higher CH_4 content and larger gas volumes can usually be achieved when they are mixed with other OM due to the low C:N ratio [66]. For example the anaerobic digestion of poultry manure is challenging due to its low C:N ratio and the accumulation of NH_3 during the process, since both free NH_3 and NH_4^+ inhibit methanogenic activity. For these reasons, co-digestion with other C-rich organic waste, such as cattle manure, crop by-products or straw have been used to improve the efficiency of anaerobic digestion [67]. The green waste generated in horticulture contains a significant amount of digestable carbohydrates and has a low buffer capacity, so the anaerobic digestion of these wastes together with animal manure is recommended [33]. Neshat et al. [68] reviewed the codigestion of animal manure and lignocellulosic residues. Lignocellulosic residues have a high C:N ratio, are available in large quantities and are inexpensive, and are therefore a good complement to animal manure in the digestion process. The mixing ratios reported varied over a fairly wide

range; establishing the optimal pH and C:N ratio is the main criterion when determining the mixing ratio.

Guilayn et al. [21] classified digestates based on the nature of the raw materials. A principal component analysis identified the first type of digestates as fibrous, consisting of cattle slurry, cattle manure and silage feedstocks. This type of digestates had a higher DM content (6.5%), a lower $\text{NH}_4\text{-N}$:total N ratio (50.7%), a lower $\text{NH}_4\text{-N}$ ($38.5 \text{ g kg}^{-1} \text{ DM}$), and a lower total N content ($75.8 \text{ g kg}^{-1} \text{ DM}$). In contrast, digestates made from typically non-fibrous feedstocks, such as pig slurry, food and agro-industrial residues, food waste, and the organic fraction of MSW feedstock were characterized by a lower DM content (3.3%), a higher $\text{NH}_4\text{-N}$: total N ratio (75.0%), a higher $\text{NH}_4\text{-N}$ (125 g kg^{-1}) and a higher total N content (155 g kg^{-1}). However, the C:organic N ratio, P and K content were not statistically different between the groups.

As a summary, Figure 2 presents the main characteristics of digestates made from different raw material groups and from mixed materials.



Figure 2. Main characteristics of digestates made from different raw material groups and their mixtures. Note that the properties of digestates may be influenced by other factors besides the raw materials (pretreatment, temperature, retention time).

4. Effect of the Digestion Process on the Quality of Digestates

The anaerobic digestion process can be divided into four main stages: hydrolysis, acidogenesis, acetogenesis and methanogenesis [69,70]. In the hydrolysis stage, large, complex organic molecules (such as proteins, carbohydrates, lipids) are broken down into simpler, medium-sized organic molecules (such as amino acids, sugars, fatty acids). During acidogenesis, medium-sized organic molecules are further broken down by acidogenic bacteria into lower molecular weight volatile fatty acids and other substances (e.g., CO_2 , NH_3 , H_2S). In the process of acetogenesis, acetogenic bacteria further break down volatile fatty acids into acetic acid (acetate), hydrogen (H_2) and carbon dioxide (CO_2). During methanogenesis, acetate, H_2 , and CO_2 are converted into CH_4 and CO_2 [69].

The mass of digested OM gradually decreases during the anaerobic digestion process as the OM is broken down and biogas is produced [15,71]. The largest mass loss occurs during the methanogenesis stage, when OM is converted into biogas [15], but this is usually only 3–10% of the original OM mass (sometimes exceeding 10%), so the mass of digestate is 90–97% of the mass of digested OM [72]. During anaerobic digestion, the C content of the OM is converted into CH_4 and CO_2 , which leads to a decrease in the C:N ratio of the material [73]. Depending on the feedstock composition and the design of the operating system, 20–95% of the OM is degraded during digestion. The quality of the digestate is determined by the moisture content and energy content of the raw materials, the duration of anaerobic digestion, and the technical parameters of the digestion plants [26].

During anaerobic digestion, the most important environmental factor regulating the structure and activity of the microbial community is temperature [74]. Operating a thermophilic ($\sim 55^\circ\text{C}$) plant requires a higher energy investment, but the amount of biogas produced will be higher and the digestate will be of high quality and free of pathogens. However, the digestion process is often unstable and plants operating in the higher temperature range are more expensive to maintain than mesophilic ($\sim 35^\circ\text{C}$) systems, which is why most commercial biogas plants are mesophilic [75].

Based on the moisture content of the raw materials, wet digesters are the most common. In these, the moisture content of the raw materials is more than 85%, such as sewage sludge or slurry [30]. Dry digesters are suitable for processing raw materials with a dry matter content over 15%, including food waste, grass clippings and energy crops [76]. Among the pre-treatments, recirculation has gained prominence in recent years. Returning a portion of the digestate produced from agricultural plant products to the system has a beneficial effect on both the process and the final product [38].

The hydraulic retention time, i.e., the time the substrate spends in the anaerobic digestion plant, must be long enough for the OM content of the raw material to be converted to biogas to a sufficient extent. A short hydraulic retention time with a high organic loading rate will result in a fairly large amount of undecomposed OM remaining in the digestate. However, if the organic loading rate is low but the hydraulic retention time is too long, then negligible amounts of CH_4 are produced [77].

The operating conditions in anaerobic digesters in wastewater treatment plants, such as the use of additives or pH adjustment, can increase the concentration of orthophosphate (PO_4^{3-}), which is a form of P available to plants [78]. Some technologies use chelating agents and cation exchange resins to bind soluble cations, thus increasing the amount of PO_4^{3-} in the final product, but this results in increased operating costs. Reduced pH significantly increases the concentration of PO_4^{3-} , but results in lower gas yield [31]. The pH of digestates from food waste increases proportionally with hygienization heat treatment (to 60°C or 121°C), while the $\text{NH}_3\text{-N}$ content decreases, because both CO_2 and NH_3 are removed from the digestate during heat treatment [47].

5. Effects of Digestates on Soil Properties

5.1. Effects on Soil Organic Matter (SOM) Content and Quality

Digestates contain large amounts of different organic molecules originating from the base materials and from the microbes involved in anaerobic digestion [26]. During anaerobic digestion, the majority of the OM input will be transformed into biogas. The biological stability of the residual organic molecules improves in the course of digestion, as the proportion of molecules more resistant to degradation, such as lignin, cutin, steroids and complex proteins, increases. When comparing the organic molecules derived from digestates, composts or other organic manures, De Neve et al. [79] found that digestates had the most stable OM. This was attributed by Głowacka et al. [80] and Smith et al. [81]

to the fact that applying digestate to the soil could contribute to C sequestration in the soil. The theory behind this is that the organic molecules remaining in the material after digestion are more stable and have lower energy content. Consequently, they are more resistant to degradation and may lead to a more lasting increase in soil OC stocks than unstabilized OM.

Besides the mineralization of humic substances, humification processes also take place during anaerobic digestion. In addition to biogas, humic acid precursors or humic-like substances are also produced in the digesters [41]. The aliphatic and aromatic molecules produced or remaining after digestion can be considered as potential humus precursors with high biological stability [82]. According to the results of Li et al. [83], 16.3% of humic acids and 27.0% of fulvic acids were decomposed in the course of sewage sludge anaerobic digestion. The humic acids extracted from digested sludge had more oxygen-containing functional groups, a more aromatic structure and larger molecule size compared to those extracted from raw sludge.

The application of digestate to the soil contributes to OM and N enrichment, thus enhancing humification processes [11]. After entering the soil, the OM in the digestate is transformed again by soil microorganisms and further mineralization and humification processes occur. However, the behavior and transformation processes of digestate depend on several factors, like soil characteristics, the crop being cultivated, the type and quantity of material applied, and other circumstances [84].

The effects of separately applied solid and liquid digestate fractions on a semi-natural grassland in Lithuania were evaluated by Jurgutis et al. [11]. Digestates were used in 85 and 170 kg N ha⁻¹ doses during three growing seasons, without soil tillage. Total OC and labile OC in the soil were determined from the 0–40 cm surface soil, in 10 cm layers. In treatments with the higher dose of solid digestate, the soil OC stock increased by nearly 16% during the experiment but this was not significant. The concentration of labile OC, however, increased significantly in the top 0–10 cm layer, with a 7–15% increase depending on the treatment. The proportion of total and labile C in the soil remained constant, however, irrespective of the treatment. This suggests that the total OC content of the 0–10 cm soil layer may be increased by long-term digestate application, even without tillage.

In relation to the study described above, a field trial was carried out [84] with the same doses and treatments, but on arable land and pasture instead of grassland. The liquid and solid phases of the digestate were applied separately in two consecutive years. Total OC content in the top 0–10 cm soil layer was increased by the solid digestate, whereas the liquid fraction resulted in an increment in the lower soil layers. Both digestate fractions had a positive effect on the mobile humic acid quantity, both on the arable land with crop rotation and on the pasture. An accumulation was observed in the 0–10 cm and 10–20 cm layers in the case of the arable land. Total OC accumulation was found in the top layers of the pasture sites, with a proportional downward decrease to the lower layers. This decrease was detected in the case of mobile OM fractions as well.

Grigatti [62] investigated food waste-based anaerobic digestates and observed enhanced biological stability, and thus possibly greater soil OC conservation in composted digestates over raw digestates. Soil OC quality was better in digestate treatments than after mineral fertilization.

Tang et al. [85] studied the effects of both digestate and digestate + straw. The treatments were effective in the 0–60 cm soil layers in terms of total OC, whereas the labile OC fraction only increased significantly in the 0–20 cm top layer. This can probably be explained by the intensified OM decomposition in the surface layer due to higher microbiological activity, resulting in the fragmentation of stable organic molecules to smaller molecular-weight substances in the digestate, which in itself is only thought to contain

small amounts of labile organic molecules [86]. On the other hand, after digestate + straw treatment, the concentration of semi-labile OC increased in all the tested soil layers, whereas the concentration of recalcitrant, or persistent OC was significantly higher in the 20–60 cm layers. The increment in total OC could be due not only to the C from digestate application, but also to the increased biomass in these treatments, including the roots and aboveground plant material plowed into the soil. The liquid nature of the digestate allows it to infiltrate into the deeper soil layers, increasing the OC stocks there [87]. Infiltration is also supported by the widespread root channels in the soil owing to the larger biomass [88]. Similar results were found by Levin et al. [89], who studied the effects of plant residue- and cattle manure-based digestate in an 8-year field experiment. A significant increase in soil C stock could be observed owing to both the C content of the applied digestate and the higher amount of root biomass, with no impact of tillage. The OC increment in the soil was 0.017% for every 1 t ha⁻¹ digestate applied.

Figure 3 shows the changes of organic substances during digestion and their effect on OC content of the soil.

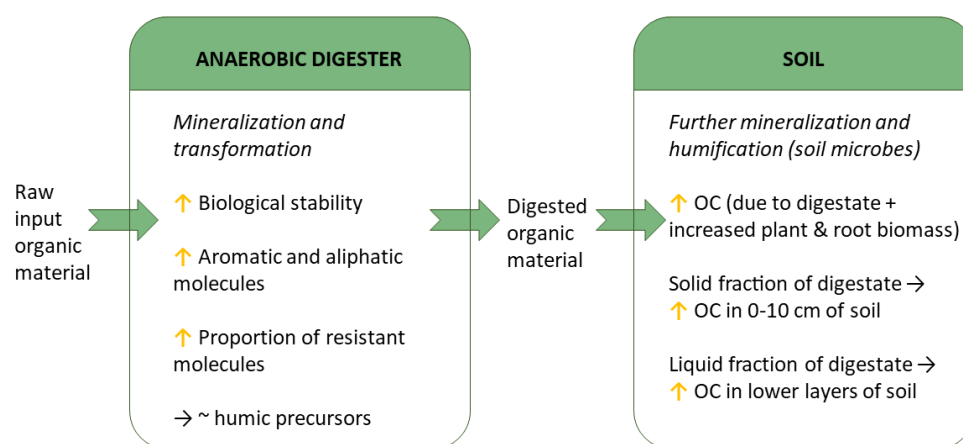


Figure 3. Transformation of organic carbon (OC) in the raw input material during digestion and its effect on soil OC content.

5.2. Effects on Element Content and Chemical Properties

5.2.1. Nitrogen Content

Mineral forms of N are easily absorbed by the plant, but the risk of losses is also high, so it is important to understand the dynamics of N mineralization after the application of digestates. In digestates, N is primarily present in the form of NH₄-N, so in the short term, their application causes the NH₄-N content of the soil to increase [43,80,90–94].

The ratio of NH₄-N to total N is an important indicator for assessing N availability [21]. In a digestate made from a mixture of food and animal manure, Rigby and Smith [63] identified 68% of the total N content as mineral N, primarily in the form of NH₄-N, which enriched the mineral N pool of the soil to a larger extent than solid waste digestate, composted food waste, or a digestate made from the organic fraction of municipal solid waste. Various digestates may contain 42–87% of total N in the form of NH₄-N [91]. Fouda et al. [95] compared seven differently prepared digestates (silage, mixture of silage and pig or cattle or dairy slurry, cattle slurry) with ammonium nitrate fertilizer at the same N dose. The digestates showed large differences in chemical composition: NH₄-N: 0.20–0.51%, total N: 0.36–0.75%, organic C: 1.85–4.75% in the fresh material. If rapid, high-dose N fertilization is required, it may be desirable to have a relatively high NH₄-N content, a high NH₄-N:total N ratio, and a low organic C:organic N ratio for the digestates. The high degree of variability in the composition of digestates also draws attention to the limitations of experiments performed with different digestates.

NH_4^+ can be converted to NH_3 , which volatilizes, so N may be lost in gaseous form. This is most significant at soil pH values of 7 or above [96]. Furthermore, within a short time after application, NH_4 is gradually converted to NO_3^- as a result of nitrification processes [41,97–99]. Depending on the composition of the digestate, 44–84% of the total N applied with digestates was detected in the soil as $\text{NO}_3\text{-N}$, after a 56-day incubation period [91].

The NH_4^+ cations applied with the digestate are absorbed on the surface of the mostly negatively charged soil particles. Nevertheless, the digestate may infiltrate into the soil, thus reaching the 20–30 cm layer [41], however, intensive leaching into deeper soil layers is not typical [43,100]. The NO_3^- anion is more mobile in the soil and therefore more prone to leaching. The rate of nitrification depends significantly on the type of digestate. According to Albuquerque et al. [91], the biodegradation of digestates is primarily determined by the biochemical oxygen demand, the amount of dissolved organic C, and the amount of organic C mineralized in the soil during the first 7 days. The ratio of these to the total N concentration determines the C and N dynamics, and ultimately the N fertilizer potential. Digestate made from mixtures of cattle slurry + glycerine is readily biodegradable, so after application, N-immobilization or denitrification processes dominate in the soil, which can lead to N deficiency in crops. In contrast, digestates based on cattle slurry + 5% orange peel waste or pig slurry + 0.6% pasteurized slaughterhouse waste have a low N immobilization rate and can provide a suitable N source for crops.

However, according to Fouda et al. [95], digestates that are more prone to initial N immobilization also increase the N content of the soil microbial biomass, and thus the total N content. After repeated application over a longer period, the positive correlation between the total N content of the soil and plant N uptake indicates that the organic N from the digestate enriches the soil N pool, thereby increasing plant N uptake.

Soil type also has an impact on nitrification processes. Nitrification occurs more rapidly in well-aerated sandy soils, which may increase N losses in the form of leaching. In clay or loam soils, the process may occur more slowly, but denitrification processes may also be more characteristic, thus reducing $\text{NO}_3\text{-N}$ concentrations and increasing volatilization losses in the form of N_2 or N_2O [63]. $\text{NO}_3\text{-N}$ leaching following nitrification can cause problems, primarily when the amount of digestate applied is excessive in terms of soil properties or crop requirements [43,90,93,101]. $\text{NO}_3\text{-N}$ leaching may be even more intensive than that of other organic fertilizers such as compost, manure or meat and bone meal [102]. However, compared to mineral N fertilizer with the same N dose, digestate application may result in similar [103] or lower NO_3^- accumulation in the soil profile [100,104–107].

5.2.2. Phosphorus, Potassium and Other Nutrient Contents

The P content of digestates can vary between 0.1 and 54 mg g^{-1} (DM) [19], with a typical value between 6 and 17 mg g^{-1} (DM) [26]. Most studies report an increase in soil-available P following the application of digestates [92,108–114]. However, the effect depends on the quality and dosage of the digestate and the characteristics of the soil. The high pH value, Mg^{2+} and Ca^{2+} content of the digestate can hinder the uptake of P, because Ca and Mg form unavailable compounds with P, such as calcium phosphates or struvite ($\text{NH}_4\text{MgPO}_4 \cdot 6\text{H}_2\text{O}$) [115,116]. Digestates produced from animal manures and energy crops generally have a low Ca:P ratio, so they can contribute more to plant P uptake, while green waste and food waste generally contain a higher Ca:P ratio [117]. High soil pH or Ca content can also hinder the plant absorption of the otherwise easily available P content of the digestate [118], but this is also the case with any other P-containing fertilizers. When applied to acidic soils, or in the case of a digestate with high Fe:P or Al:P ratio, the formation of Fe and Al hydroxides may hinder the uptake of the applied P and enrich the total P content of the soil [117].

Several studies also report an increase in soil K content due to digestate treatment [71,80,98,106,109,112,119–121]. However, the extent of the increase in K content also depends on the application rate and on the K content of the digestate, which typically ranges from 19 to 43 mg g⁻¹ (DM) [26], though it may also range from 1 to 110 mg g⁻¹ (DM) [19]. Regarding the raw materials used, Brtnicky et al. [122] recommend the use of legumes, which can increase the K content of the digestate. According to Makádi et al. [98], the increase in K also depends on the soil type: on sandy soil with lower K supply, the K₂O concentration increased to a greater extent (from 87 to 165 mg kg⁻¹), while on loamy soil with a higher K supply, the K₂O concentration increased to a lesser extent (from 150 to 189 mg kg⁻¹) under the influence of the same dose of digestate treatment. In some experiments, there was no significant change in soil K concentration [93,102].

In addition to macroelements, applying digestate can also increase the concentration of meso- and microelements such as Mg, Ca, [113,121,123], Cu, Zn, Mn [80,93,121,124] and Fe [109,121] in the soil. However, these elements are generally present in lower concentrations in digestates, so significant changes can typically be detected only at higher application rates [80]. It is also possible that the total soil content of some elements increases, while their plant-available amount decreases. The decrease in the observed available fraction of Mn is likely due to the fact that digestate greatly increased the sorption capacity of the soil [80].

The extent of the change also depends on the properties of the soil. Compared to coarser sandy soils, loamy and clay soils contain a higher proportion of clay minerals with a larger specific surface area and more binding sites for cations [125]. The pH change caused by digestates may affect the solubility of nutrients, the ratio of total to available fraction either positively or negatively [121]. Valentinuzzi et al. [110] highlighted that the exchangeable cation content (Ca, Mg, K and Na) of digestate can modify the exchangeable cation content and ratio of the soil, so digestate treatment may reduce the Ca:Mg and Mg:K ratios, which may also affect the solubility of the elements.

The application of digestate may not be immediately associated with changes in the soil. In a 4-year field experiment, Odlare et al. [126] found that digestate did not change the chemical properties of the soil. Nevertheless, with a few exceptions, it can generally be concluded that the short-term effects of digestate application have a positive impact on soil quality indicators [27].

Table 1 summarizes the percentage changes in the elemental composition of soil caused by different types of digestate.

Table 1. Impact of digestates on changes in soil element contents.

Element (Form)	Change of Content	Soil	Raw Materials of Digestate	Digestate Rate	Reference
N (total)	5.4–45% increase	Silt loam	Swine manure	Corresponding to 170–680 kg N ha ⁻¹	[124]
	7% decrease–16% increase	Silty clay	Energy crops, agricultural by-products, cattle slurry	34,700 L ha ⁻¹	[127]
	41–55% increase	Sandy loam	Cattle and pig manure	Corresponding to 2.56 g N per plant	[111]
	39–58% increase	Sand	Cow and poultry manure, plant residues, other organic residues, sterilized animal residues	Corresponding to 170 kg N ha ⁻¹	[98]
	6.4–13% increase	Loam	Cow and poultry manure, plant residues, other organic residues, sterilized animal residues	Corresponding to 170 kg N ha ⁻¹	[98]
	14–19% increase	Silty loam	Corn silage, sugar bagasse beet, pomace of fruit, dairy waste, manure	30–60 m ³ ha ⁻¹	[80]
(available)	52–61% increase	Sandy loam	Cattle and pig manure	Corresponding to 2.56 g N per plant	[111]

Table 1. Cont.

Element (Form)	Change of Content	Soil	Raw Materials of Digestate	Digestate Rate	Reference
P (total) (available)	11% decrease–62% increase	Silt loam	Swine manure	Corresponding to 170–680 kg N ha ^{−1}	[124]
	19% increase	Texture not specified	Green waste, maize silage, beet pulp, stillage, whey	36,000 L ha ^{−1}	[113]
	12–20% increase	Sandy loam	Cattle and pig manure	Corresponding to 55.8 g P per plant	[111]
	212–218% increase	Sandy loam	Cattle and pig manure	Corresponding to 55.8 g P per plant	[111]
	50–54% increase	Silty loam	Corn silage, sugar bagasse beet, pomace of fruit, dairy waste, manure	30–60 m ³ ha ^{−1}	[80]
	26–44% increase	Sand	Cow and poultry manure, plant residues, other organic residues, sterilized animal residues	Corresponding to 170 kg N ha ^{−1}	[98]
K (total) (exchangeable) (available)	10–34% increase	Loam	Cow and poultry manure, plant residues, other organic residues, sterilized animal residues	Corresponding to 170 kg N ha ^{−1}	[98]
	2.4–17% increase	Silt loam	Swine manure	Corresponding to 170–680 kg N ha ^{−1}	[124]
	128% increase	Texture not specified	Green waste, maize silage, beet pulp, stillage, whey	36,000 L ha ^{−1}	[113]
	4% decrease–22% increase	Silty clay	Energy crops, agricultural by-products, cattle slurry	34,700 L ha ^{−1}	[127]
	0–36% increase	Loamy sand	Sugar beet and maize in different ratios	30 m ³ ha ^{−1}	[125]
	3.8 decrease–62% increase	Sand	Sugar beet and maize in different ratios	30 m ³ ha ^{−1}	[125]
	59–67% increase	Sandy loam	Cattle and pig manure	Corresponding to 7.56 g K per plant	[111]
	39–73% increase	Silty loam	Corn silage, sugar bagasse beet, pomace of fruit, dairy waste, manure	30–60 m ³ ha ^{−1}	[80]
	25–94% increase	Texture not specified	Corn silage, straw, chicken manure, pig slurry, distiller's grain	Corresponding to 28–112 mg N kg ^{−1}	[121]
	49–90% increase	Sand	Cow and poultry manure, plant residues, other organic residues, sterilized animal residues	Corresponding to 170 kg N ha ^{−1}	[98]
Mg (total) (exchangeable) (available)	1% decrease–26% increase	Loam	Cow and poultry manure, plant residues, other organic residues, sterilized animal residues	Corresponding to 170 kg N ha ^{−1}	[98]
	3% increase	Texture not specified	Green waste, maize silage, beet pulp, stillage, whey	36,000 L ha ^{−1}	[113]
	4% decrease–13% increase	Silty clay	Energy crops, agricultural by-products, cattle slurry	34,700 L ha ^{−1}	[127]
	0–19% increase	Loamy sand	Sugar beet and maize in different ratios	30 m ³ ha ^{−1}	[125]
	4.4% decrease–20% increase	Sand	Sugar beet and maize in different ratios	30 m ³ ha ^{−1}	[125]
Zn (exchangeable)	47–53% increase	Silty loam	Corn silage, sugar bagasse beet, pomace of fruit, dairy waste, manure	30–60 m ³ ha ^{−1}	[80]
(available)	11% decrease–16% increase	Silty clay	Energy crops, agricultural by-products, cattle slurry	34,700 L ha ^{−1}	[127]
	3.6–32% increase	Silty loam	Corn silage, sugar bagasse beet, pomace of fruit, dairy waste, manure	30–60 m ³ ha ^{−1}	[80]
Ca (exchangeable)	5% decrease–2% increase	Silty clay	Energy crops, agricultural by-products, cattle slurry	34,700 L ha ^{−1}	[127]
	6.3% decrease–2.2% increase	Loamy sand	Sugar beet and maize in different ratios	30 m ³ ha ^{−1}	[125]
	1.6–28% increase	Sand	Sugar beet and maize in different ratios	30 m ³ ha ^{−1}	[125]
Cu (available)	9% decrease–4% increase	Silty clay	Energy crops, agricultural by-products, cattle slurry	34,700 L ha ^{−1}	[127]
	7.1% decrease–36% increase	Silty loam	Corn silage, sugar bagasse beet, pomace of fruit, dairy waste, manure	30–60 m ³ ha ^{−1}	[80]
Fe (available)	9% decrease–2% increase	Silty clay	Energy crops, agricultural by-products, cattle slurry	34,700 L ha ^{−1}	[127]
	2.9% decrease–1.2% increase	Silty loam	Corn silage, sugar bagasse beet, pomace of fruit, dairy waste, manure	30–60 m ³ ha ^{−1}	[80]
Mn (available)	23% decrease–15% increase	Silty clay	Energy crops, agricultural by-products, cattle slurry	34,700 L ha ^{−1}	[127]
	9.5–12% decrease	Silty loam	Corn silage, sugar bagasse beet, pomace of fruit, dairy waste, manure	30–60 m ³ ha ^{−1}	[80]

5.2.3. Electrical Conductivity—Salinity

Electrical conductivity (EC) is an excellent indicator of salinity, although it cannot distinguish between dissolved nutrients and salts that are harmful to the soil. Nevertheless, a high value above 2 mS cm^{-1} suggests the need for intervention [128]. During anaerobic digestion, OM decays and the proportion of minerals increases. This process increases the concentration of soluble salts in the digestate [129,130]. The generally high salt content of digestates directly increases the electrical conductivity of the soil [33,121,123]. The extent of the change depends on the salinity of the digestate, the application rate, the soil properties and environmental conditions. In some experiments, the EC value exhibited a several times increase [99,124], in other studies a significant increase was observed compared to the control [46,110,123,131] or to manure treatment [120]. It is important to note that the dissolved N, primarily the $\text{NH}_4\text{-N}$ form, as well as the P, K, Ca and Mg ions in digestates can also significantly contribute to an increase in soil EC value [92,123], while Ca may increase the salt tolerance of crops, reducing the effects caused by high EC [132].

Digestates made from various raw materials may have different mineral contents [110,129]. Food-based digestates often contain a high proportion of Na, even in the order of g kg^{-1} , because the digested food waste itself is also characterized by a high salt content. The application of such digestates to soil may be associated with the risk of salinization, i.e., deterioration of the physical and chemical properties of the soil, and a decrease in fertility [133]. In a field experiment, Pawlett and Tibbett [46] applied food-based digestate containing 1671 mg kg^{-1} Na at 100 and 200 kg N ha^{-1} doses, 7 times over 2 years. The results showed significant increases in the amount of available Na in the soil and in the sodium adsorption ratio (SAR) after digestate treatments. This raises a potential risk for land areas that are continuously exposed to similarly high-salt digestates, especially if the soil in the area is prone to salinization. According to studies by Meng et al. [123], the total Cl^- ion content in the soil was greatly influenced by the concentration of food waste digestate being 205 mg kg^{-1} at 10% dilution, 320 mg kg^{-1} at 50% dilution, and 299 to 369 mg kg^{-1} at 100% concentration.

The ratio of salt content to N content in digestates is also an important factor. Jamison et al. [133] found that EC of the growing medium increased significantly when exposed to lignocellulosic-biomass digestate from energy crops with an EC of 8.34 mS cm^{-1} , whereas exposure to food-waste digestate with an EC of 10.99 mS cm^{-1} did not result in a significant change. This could be attributed to the fact that the lignocellulosic-biomass digestate had a much lower N content, so a higher dose was required to achieve the same amount of N, thus applying a much higher amount of salt than with the smaller dose of food-waste digestate.

Soil texture also affects the rate of change in EC. Voelkner et al. [130] found a significant increase in EC after the addition of anaerobic digestate in both loamy and sandy soils with slightly acid pH. However, on loamy soil with originally higher EC, the treatment resulted in higher EC values due to the larger specific surface area, because a higher proportion of ions was adsorbed on the soil particles. Furthermore, the mechanical pre-treatment of the input substrates also affected the rise in EC as a function of soil texture. Grinding the digestate raw materials resulted in a larger specific surface area, faster degradation, and greater transfer of electrolytes, which increased the EC of loamy soil to a greater extent than that of sandy soil. Chopping, on the other hand, resulted in a digestate with higher dry matter content and fiber length, which was better able to penetrate into the pores of sandy soil, thus having a stronger effect on EC in this soil.

Digestates have an interactive effect on soil EC and soil microbial activity. After the application of digestate, the soil EC may change over time due to microbial activity and mineralization processes [131]. However, digestates with a high EC value can have a detrimental effect on soil microbial activity [130,134].

When examining the effect of digestates on soil EC, the effect of mineral fertilization on EC must also be taken into account. Yu et al. [111] showed in a field study that EC clearly increased in both digestate and fertilizer treatments compared to the control. In a study by Albuquerque et al. [92], mineral fertilization resulted in higher EC values than digestate treatment with the same N, P and K active agent content.

In arid regions or during dry periods, the application of digestates with high salt content enhances the accumulation of salts in the soil [92]. Even shorter wet or dry cycles can affect EC [41]. However, with adequate irrigation, even digestates with a high EC value do not cause an increase in EC in the soil, and may even result in a decrease [135]. The proper dilution and dosage of digestates with high salt content is key to minimizing negative impacts on the soil [127].

5.2.4. pH

The effect of digestate on soil pH is important for plant nutrient uptake, as pH significantly influences the solubility and availability of nutrients [15,80]. The optimal soil pH value for crop nutrient absorption is between 6 and 7, but this can vary depending on the plant species [136]. Digestate treatment may cause changes in soil pH for several reasons. In general, the pH of digestate is neutral or slightly alkaline [98,134], possibly leading to an increase in soil pH, especially in acidic soils [11,80,109,113,121,137,138].

Since soil acidification affects large areas, the pH-increasing effect of digestates can be particularly beneficial on acidic soils [93]. Liu et al. [138] observed that replacing 75% of the fertilizer dose with digestate significantly increased soil pH from 5.63 to 5.86 using digestate with a pH of 7.62. Meng et al. [123] found that digestate prepared from food waste had a pH of 9.02, so higher input raised the soil pH from 6.30 to 7.02.

However, the effect of digestate on soil pH is not always clear, and in some cases no significant change was observed [98,110], or a decrease may occur [98,124,125]. The effect depends on the pH value of the digestate, the application rate and the specific properties of the soil [134]. The relatively high NH_4^+ content of digestate may increase soil pH [129,130], but the H^+ ions released during the nitrification of NH_4^+ could acidify the soil [116]. Jin et al. [124] found that although digestate was slightly alkaline, soil pH decreased after application, which was attributed to the nitrification of NH_4^+ . Kataki et al. [109] also found that digestates with relatively higher NH_4^+ content were more likely to reduce soil pH due to nitrification.

Digestate may contain alkaline cations such as Na^+ , K^+ , Ca^{2+} and Mg^{2+} , which can alter the cation balance on soil colloids and increase pH [125,131]. During anaerobic digestion, alkalinizing agents or carbonates may be added to the system to control pH, possibly causing a further increase in the pH of the digestate [129]. According to Bougnom et al. [120], the pH of slurry digestates is around 8, which is approximately 0.5 units higher than that of undigested slurry. The dried or separated solid part of the digestate may have more alkaline pH than the liquid fraction [133,137].

Due to incomplete anaerobic digestion, the content of volatile fatty acids may also be significant, thus reducing the pH value of the digestate [131]. Makádi et al. [98] observed that treating slightly acidic sandy soil with digestate resulted in a slight decrease in pH, which could be due to the presence of acidic compounds in the digestate. Voelkner et al. [131] observed a decrease in soil pH after the addition of digestate to loamy soil, possibly caused by the organic acids formed during the anaerobic fermentation of the digestate and the microbial oxidation of OM in the digestate. However, according to Jin et al. [124], the application of mineral fertilizers caused a more significant pH decrease than that of digestates. Moreover, the degradation of the volatile fatty acids in the soil can potentially lead to an increase in pH over time [109,129].

Soil pH and buffer capacity have a significant influence on the extent to which digestate application changes the pH [121,131,134]. Soils with low buffer capacity are more sensitive to the pH of the digestate [98]. In contrast, soils with high buffer capacity, with high cation exchange capacity (CEC) and base saturation, may be more resistant to pH changes [92]. Furthermore, the increase in pH is smaller in soils with a higher pH. Koszel and Lorencowicz [113] therefore detected only a slight increase in soil pH from 7.56 to 7.63 after applying digestate with a pH of 8.73. Alburquerque et al. [92] also found that digestate with a pH of 8.3 caused only minimal changes to calcareous soil with a pH of 7.9 due to the small pH difference and the high buffering capacity of the calcareous soil. Table 2 summarizes the effects of digestates on soil pH.

Table 2. Selected references for the change in soil pH due to digestate application.

pH Change	Raw Materials of Digestate	Digestate pH	Change in Soil pH Compared to Control Treatment (as a Result of the Applied Digestate Rate)	Reference
pH increased	Chicken manure	7.62	pH(H ₂ O) from 5.63 to 5.71 (digestate replacing 25% N fertilizer: 27,167 kg ha ⁻¹ digestate)–5.86 (digestate replacing 75% N fertilizer: 81,333 kg ha ⁻¹ digestate)	[138]
	70% corn silage, 15% sugar bagasse beet, 5% pomace of fruit, 5% dairy waste, 5% manure	7.8–8.5	pH(KCl) from 4.5 to 4.6 (30 m ³ ha ⁻¹ digestate) and 4.9 (60 m ³ ha ⁻¹ digestate)	[80]
	Diluted chicken manure, food waste, maize and beetroot waste	8.1–8.74 (solid) 7.94–8.17 (liquid)	pH(KCl) from 7.68 to 7.82 (solid or liquid digestate corresponding to 85 kg N ha ⁻¹) and 7.77 (liquid digestate) and 7.80 (solid digestate corresp. to 170 kg N ha ⁻¹)	[11]
	Green waste, maize silage, beet pulp, stillage, whey	8.73	from 7.56 to 7.63 (36,000 L ha ⁻¹ digestate)	[113]
	Liquid: pig slurry, corn-based distiller's grain, cattle slurry Solid: corn silage, straw, chicken manure	7.90 (liquid) 8.25 (solid)	pH(KCl) from 4.52 to 4.67 (liquid digestate corresponding to 112 kg N ha ⁻¹) and 4.62 (solid digestate corresponding to 84 kg N ha ⁻¹)	[121]
	95% food waste (restaurant), 5% gardening waste	6.82 (liquid) 8.86 (dried)	pH(KCl) in winter/spring season from 4.73 to 4.98 (dried) and 5.46 (liquid) in spring/summer season from 4.88 to 5.36 (dried) and 5.56 (liquid) (corresponding to 140 kg N ha ⁻¹)	[137]
	Cow dung (CD) or <i>Ipomoea carnea</i> leaves + cow dung (ICD) or Rice straw + green gram stover + cow dung (RGC)	7.3 (CD), 7.8 (ICD), 8.0 (RGC)	pH(H ₂ O) from 5.43 to 5.60 (CD), 5.48 (ICD), 5.51 (RGC) (digestate corresponding to 60 kg N ha ⁻¹)	[109]
pH decreased	M100: maize or SB100: sugar beet or SB80M20: 80% sugar beet and 20% maize or M80SB20: 80% maize and 20% sugar beet	M100: 8.25–7.93 SB100: 8.21–7.85 SB80M20: 7.74–7.82 M80SB20: 7.75–7.85	pH(CaCl ₂) loamy soil from 6.64 to 7.24–7.24 (M100); 7.23–7.28 (SB100); 7.16–7.23 (SB80M20); 7.16–7.21 (M80SB20) sandy soil from 5.19 to 5.74–5.82 (M100); 5.65–5.71 (SB100); 5.76–5.83 (SB80M20); 5.81–5.89 (M80SB20) (30 m ³ ha ⁻¹ digestate)	[130]
	Not defined	7.5	pH(CaCl ₂) from 5.78 to 5.69 (digestate corresponding to 190 kg N ha ⁻¹)	[120]
	Swine manure	For eggplant: 8.1 (solid–SD) 8.3 (liquid–LD) For cabbage: 8.1 (SD) 9.2 (LD)	pH(H ₂ O) Eggplant: from 6.99 to 6.62 (13,375 kg ha ⁻¹ SD + 90,785 kg ha ⁻¹ LD)–6.36 (53,500 kg ha ⁻¹ SD + 224,800 kg ha ⁻¹ LD) Shanghai cabbage: from 7.15 to 6.90 (13,375 kg ha ⁻¹ SD + 90,785 kg ha ⁻¹ LD)–6.65 (53,500 kg ha ⁻¹ SD + 224,800 kg ha ⁻¹ LD)	[124]
pH both increased and decreased	42.4% animal residues, 22.3% cow manure, 18.9% organic residues, 12.0% plant residues, 4.4% poultry manure	8.03	1st year: sandy soil: from 6.27 to 6.29; loamy soil: no change (8.13); 2nd year: sandy soil: from 6.40 to 6.25; loamy soil: from 7.64 to 7.67 (digestate corresp. to 170 kg N ha ⁻¹ year ⁻¹)	[98]

5.2.5. Cation Exchange Capacity

The CEC value of soils is fundamentally determined by the negative charges on the surface of clay minerals and SOM, which enable the soil to bind cations. OM has a high CEC value; therefore, digestates with a high OM content can increase the OM content of the soil [112,134], and thus, in theory, the CEC of the soil [80]. However, although quite a few studies report CEC results, they do not give a clear, consistent confirmation of an increase in soil CEC values. In the case of CEC, too, the effect depends on the properties of the soil and digestate.

Głowacka et al. [80] found a significant increase in CEC following digestate treatment, but NPK mineral fertilization also resulted in a similar increase in CEC in a three-year field experiment. Pastorelli et al. [127] found no difference due to digestate treatment in a three-year field experiment, which was explained partly by the fact that the total organic carbon (OC) content did not change significantly, and partly by the fact that OM generally has less effect on CEC in the silty-clay texture soil used than in coarse-textured soils. This was confirmed by the results of Voelkner et al. [125], who found that both digestate and soil quality determined the impact on CEC. Digestates prepared by monofermentation (100% maize or 100% sugar beet) increased the CEC value of loamy soil, while digestates prepared by cofermentation (maize + sugar beet) decreased it, while a uniform increase was observed on sandy soil. In addition to soil texture, pH can also have an influence on CEC, as CEC may increase with rising pH due to the dissociation of protons in pH-dependent charges [139]. However, this phenomenon was observed by Voelkner et al. [125] only in the cofermentation treatment on loamy soil, where pH and CEC decreased in parallel, while the increase in CEC on sandy soil was not accompanied by a clear increase in pH. The digestate from monofermentation resulted in lower pH and higher CEC values. This may indicate that regardless of the pH change, the OM content of the digestate from monofermentation could have a beneficial effect on CEC by increasing negatively charged surfaces. In a study by Valentinuzzi et al. [110], there was no consistent effect, as CEC increased with the highest dose of liquid digestate in the case of maize test plants, but decreased with the lower dose and at both treatment rates in the case of cucumber test plants. Rolka et al. [121] generally found a decrease in CEC values in digestate-treated soils, but the extent of the decrease was mostly not significant and was not proportional to the digestate dose applied.

5.3. Effects on Physical Soil Properties

The effect of digestates on physical soil properties show significant variability depending on the type and composition, primarily the OM content, of the digestate and the soil properties of the experimental site. The dry matter content of digestates is typically 4–10%, providing 6–12 kg of organic C per ton [41].

Changes in soil texture require significant organic matter accumulation, and the stability of organic matter determines whether these changes will be permanent or temporary [69,93]. Depending on the quality of organic matters, digestates can even have the opposite effect on soil water management. Digestates can increase the water repellency of soil if they are rich in certain organic compounds, such as amphiphilic substances, long-chain fatty acids and humic substances. This is because, when the moisture content of the soil decreases below a certain level, these hydrophobic, non-polar groups turn into the pore space. This reduces the wettability of the soil and makes the surface water-repellent. Sandy soils are generally more sensitive to hydrophobization caused by digestate than soils with higher clay content because sand particles with lower specific surface areas are more quickly covered by amphiphilic organic molecules. Increased water repellency can reduce water infiltration into the soil, which can increase the risk of soil erosion and, in the short term, reduce soil fertility due to nutrients lost through surface runoff [130,131].

On the other hand, digestate, especially the solid fraction, has a high content of total organic carbon (TOC), which can stabilize the soil structure by forming organo-mineral complexes, improve aggregate stability and porosity, thus reducing soil bulk density [41,112,121,127]. Macropores help the soil aerate and allow water to infiltrate, while micropores increase its water-holding capacity. In clay soils, interactions between clay minerals and organic matter can lead to a greater formation of stable aggregates. In clay-rich soils, the formation of stable aggregates can be more intensive through the interactions between clay minerals and organic matter [125,131]. However, improvements in bulk density and porosity are not always observed in the short term, and some tillage practices, such as plowing, may offset these positive effects [127]. The application of digestate at a rate of 15 t ha⁻¹ significantly reduced the bulk density of the surface layer of sandy loam soil, which resulted in better porosity, i.e., aeration and water infiltration, and increased the water holding capacity of the soil [140]. Singh et al. [119] reported similar results and an increase in the stability of soil aggregates. El-Bakhshwan et al. [141] obtained the opposite result in clay loam soil, i.e., cumulative infiltration decreased with increasing application rate of digestate, as it promoted the swelling of soil aggregates. However, the authors did not evaluate the decrease in infiltration as a negative process, because it may improve water use efficiency by limiting percolation losses to deeper layers, thus keeping water available for plants in the root zone for a longer period.

Some authors have reported the contradictory effects of digestates on bulk density after the application of digestate. According to Jaša et al. [142], bulk density increased leading to a decrease in porosity and minimal air capacity. Penetrometer testing of the soil showed increasing compaction during the treatment, compared to the control on sandy clay soil (light texture Cambisol). In another study, however, the application of digestate was found to reduce the bulk density of the soil and increase its porosity in sandy loam [143]. It was also found to increase aggregate stability in silt loam and sandy loam [144]. According to El-Bakhshawan et al. [141], in sandy clay loam soil, the penetration resistance was reduced by the digestate, which could improve root growth and root distribution. However, Pastorelli et al. [127] found no significant difference in the bulk density of silty clay soil, while the beneficial effect of digestate on aggregate stability disappeared the following year.

Digestates can increase the amount of macroaggregates (>5 mm) in the soil, while simultaneously reducing that of microaggregates, and can also increase aggregate stability, thus improving the erosion resistance of soils [145]. After 6 years of digestate application, the physical properties and structure of the soil improved, while its bulk density decreased significantly and its water-holding capacity increased. The average diameter of the water-stable aggregates also increased, thus improving the supply of water, air and nutrients [146]. However, according to the results of Dai et al. [147], aggregate formation, aggregate stability, aggregate diameter, and the proportion of large macroaggregates were lower in the digestate treatment than in straw treatment on sandy soil, while the amount of C applied with straw was twice the amount applied with digestate. Koch et al. [148] found contradictory results on sandy soil, when examining the effect of digestate in a long-term experiment. The proportion of macropores and the hydrophobicity increased in the topsoil, while bulk density decreased. The water-holding capacity was not affected by the treatment. The increase in soil water repellency predicts that surface runoff may occur during significant rainfall events, while in sandy soils the increase in macroporosity likely increases the saturated hydraulic conductivity due to increased pore connectivity and a stabilized pore system [148].

The changes caused in the soil by the digestate also affect its cultivability. Digestate treatment reduced the plasticity index and the resistance of clay soil to tillage by 1 horsepower [149]. On sandy loam soil, the total porosity of the soil tended to increase

as a result of digestate treatment, while the bulk density did not change [150]. Voelkner et al. [125] investigated the effect of digestates on physical soil properties in sandy and loamy soils, using the readily dispersible clay content (RDC) as an indicator. The loamy soil, with higher clay and RDC content, reacted more sensitively to the effects of the dispersive agents in digestates compared to sandy soil. RDC proved to be a suitable indicator for evaluating the sensitivity of soils with different structures to dispersion. By examining the salt content of the digestates, it was determined that Na made the greatest contribution to the dispersion of soil particles. After digestate treatment, the loam soil became acidic and the sandy soil became alkaline. More acidic conditions reduced the dispersion, while more alkaline conditions increased the amount of RDC, possibly due to a change in the charge of the clay particles. The most frequently detected changes in physical soil parameters are summarized in Table 3.

Table 3. Impact of digestates on physical soil properties.

Parameter	Change	Soil Texture	Raw Materials of Digestate	Digestate Rate	Reference
Water-holding capacity	Increased	Loamy soil	Pig manure + cleansing water	120 m ³ ha ⁻¹	[146]
		Sandy loam	Agricultural waste	15 t ha ⁻¹	[140]
	No change	Sandy soil	Dairy slurry, maize silage, wheat grain	30 m ³ ha ⁻¹ year ⁻¹	[148]
Bulk density	Decreased	Sandy loam	Cattle dung	Corresponding to 150 kg ha ⁻¹ N	[143]
		Loamy soil	Pig manure + cleansing water	120 m ³ ha ⁻¹	[146]
		Sandy soil	Dairy slurry, maize silage, wheat grain	30 m ³ ha ⁻¹ year ⁻¹	[148]
		Silt loam and sandy loam	Corn silage, pig slurry, farmyard manure, hay	Corresponding to 120–150 kg ha ⁻¹ N	[144]
		Silty clay loam	Pig manure and urine	Corresponding to 18–120 kg ha ⁻¹ N	[145]
	No change	Sandy loam	Agricultural waste	4 and 15 t ha ⁻¹	[140]
		Silty clay	Energy crops, agricultural by-products, cattle slurry	34,700 L ha ⁻¹	[127]
		Sandy loam	Cow dung and brewery factory residue	Up to 61.8 m ³ ha ⁻¹	[150]
		Sandy clay	Agricultural by-products	20–40 t ha ⁻¹	[142]
	Increased	Sandy clay	Agricultural by-products	20–40 t ha ⁻¹	[142]
Aggregate stability	Increased	Silt loam and sandy loam	Digestate I: corn silage, cattle slurry; Digestate II: corn silage, pig slurry, farmyard manure, hay	Corresponding to 120–150 kg ha ⁻¹ N	[144]
		Silty clay	Energy crops, agricultural by-products, cattle slurry	34,700 L ha ⁻¹	[127]
		Silty clay loam	Pig manure and urine	Corresponding to 36–54 kg ha ⁻¹ N	[145]
		Loamy soil	Pig manure + cleansing water	120 m ³ ha ⁻¹	[146]

5.4. Effects on the Biological Properties of Soils

Digestate application has both indirect and direct effects on the quantity of SOM. The high N content of the OM applied with digestate leads to enhanced microbial activity. Microbial necromass is a highly important contributor to the recalcitrant and stable organic fraction of the soil. Necromass-derived organic molecules are more resistant to degradation, because of their strong sorption to soil minerals [151]. The meta-analysis of Wang et al. [152] revealed that globally, 35–51% of the total OC stock originates from microorganisms.

Chen et al. [153] carried out a 12-year digestate application experiment on rice fields. Their results showed that OM from the dead microbe population in long-term treatments contributed significantly to the OM stock of the soil. By promoting the permanent C pool of the soil, digestate may also decrease the risk of OM loss. Dead fungal biomass played a larger role in this process than bacterial biomass. The labile C-fractions of the substrate used for anaerobic digestion are already degraded during the digestion process, therefore, lower microbiological activity can be observed in the soil after their application than if the original organic material had been applied without digestion [154]. The proportion of the remaining C, being more resistant to degradation, is higher in the digestates than in the original substrates [155]. Depending on the input material, this more stable fraction may

have different properties, so the stability of the digestate added to the soil also varies [91]. Moreover, digestates also contain microbial biomass, but its composition is different from that of both the substrate and the soil, too and it can serve as precursor for recalcitrant organic fraction in soil [156,157].

The favorable pH range for microbial activity is between 5.3 and 9.0. This should be considered when applying digestate [123]. The C:N ratio is another important characteristic of the digestate as it determines the subsequent transformation processes when it is applied to soil. Higher OM content in the input materials and a shorter digestion time result in a higher C:N ratio and vice versa [93,158]. The low C and high readily-available N content of digestates can lead to a priming effect in the soil, which may also result in the decomposition of the original OM of the soil [91,159,160].

Rigby and Smith [63] observed a decrease in the N content of microbial biomass after digestate application to a silty clay soil. The digestate was made from food waste and animal slurry. Although a high content of $\text{NH}_4\text{-N}$ in the digestate may inhibit soil microorganisms, in this case the NH_4^+ toxicity was only temporary and decreased with nitrification activity. At the end of the incubation period, the N content of the soil microbial biomass returned to the original level. Abubaker et al. [71] compared the effects of mineral fertilizer and different digestates from distiller's waste from ethanol production, silage from ley and organic household waste, slaughterhouse waste and organic household waste. During the experiment, both N mineralization capacity and potential ammonium oxidation rate increased in nearly all the treatments, digestates having a stronger increasing effect, than mineral NPK fertilization.

Ren et al. [97] compared food waste digestate and a synthetic liquid urea ammonium nitrate solution in terms of plant growth and certain bacterial and fungal characteristics of the soil. Digestate not only stimulated plant growth to the same extent as the synthetic amendment, but it also increased the density of arbuscular mycorrhizal fungal hyphae. The application of digestate was recommended over conventional fertilizers, because not only may it have less influence on the diversity and function of soil microflora compared to conventional fertilizers, but it can also be a source of OC and nutrients in the soil.

The experiments described in the review paper of Karimi et al. [23] revealed the positive impact of digestate application on soil biological parameters. However, 7% of the studies reported the opposite effect, suggesting the need for further investigations in this field. In terms of fungal abundancy and overall microbial activity, lower values were observed in some cases when digestate was applied, indicating that digested OM had poorer quality than the original material. Digestate proved to stimulate microorganisms better than fertilizers according to nearly all the 28% of parameters that showed differences between the two materials. Compared to other organic fertilizers, digestates stimulated microbial communities more stronger. These results suggest that using digestates instead of fertilizers is beneficial.

In a laboratory incubation experiment, Stumpe et al. [161] tested the effects of digestate, liquid manure and waste-activated sludge on the soil C stock and microbial activity in sandy and loamy soils. At first, respiration values proved to be the highest in the case of the digestate, suggesting that it contained the most easily decomposable OM. In contrast, the highest concentration of water-soluble OC was found not in the digestate, but in the liquid manure. The high respiration value after digestate application was due to the mineralization of the original carbon stock of both soils. The priming effect was the strongest in liquid manure treatments. Based on these results, digestate generates similar changes in soil C stocks and microbial activity to the other organic manures tested.

Odlare et al. [126] observed improved soil biological activity and increased N mineralization after digestate addition in a four-year field experiment. Sapp et al. [162] evaluated

the effects of digestate application in terms of soil microorganism structure and diversity in an arable land with spring wheat. Digestate and mineral fertilizer caused similar improvements in wheat yield. Forty different bacterial strains were detected in the soils, with the highest proportion of *Proteobacteria*, *Acidobacteria* and *Actinobacteria*, while there was a decrease in *Planctomycetes*, which play a role in the N cycle, so digestates can promote microbial growth and plant development.

Muscolo et al. [69] aimed to find the optimal combination of plant and animal wastes as feedstocks for anaerobic digestion, focusing on their effects on soil microorganisms and enzymes. Two digestates were tested, one (F) produced from animal manure (60%), whey, corn silage, olive residue (20%) and citrus pulp (20%), and the other (U) from olive residue (30%), citrus pulp (30%) and 40% animal manure and corn silage. Compared to the control, both digestates resulted in an increase not only in the SOM content, but also in the C:N ratio and the phenol content of the test soil. A significant positive change was detected in fluorescein-diacetate and catalase activity after the application of F digestate, whereas only the highest concentration the U digestate had a positive effect on catalase activity. The results suggest that the quality of the feedstock material used for digestate production had a considerable influence on the microbial parameters analyzed, and thus on some soil properties as well.

Besides its impacts on microbiological activity, digestates affect also larger soil living organisms, such as nematodes [163]. Westphal et al. [164] studied the effects of digestate produced from the mesophilic fermentation of whole corn plants on the test plant *Beta vulgaris altissima*. Three and five weeks after planting, decreased *Heterodera schachtii* nematode infection and improved plant growth were observed, suggesting that the digestate had nematode inhibiting properties. Further investigations will be required to make utilization more effective. Moinard et al. [165] investigated how applying anaerobic digestate to the soil affects earthworms on a timescale of a few hours to two years. They detected short-term toxicity (mortality via direct contact on the surface 20 min after spreading), but neutral to positive effects on the earthworm population in the longer term, probably due to the OM added with the digestate. The short-term toxicity was found to depend on the concentration of NH_3 .

The impact of anaerobic digestate application on soil life was evaluated in the review paper of van Midden et al. [24]. The emphasis was placed on the importance of the quantity and composition of digestate C. The effect of these materials on the soil macrobiota can be categorized according to the habitat. The impact on surface-dwelling organisms, such as earthworms, springtails and nematodes, is more harmful than on soil-dwelling organisms, possibly due to NH_3 toxicity. Regarding the C content of the digestate, the higher it is, the more positive impact it has on microbial abundance and diversity. There is also a difference regarding the availability of C. While labile C is more favorable for fast-growing bacteria, recalcitrant C stimulates slower-growing fungi. As different feedstocks and production processes result in digestates with varying properties, and because repeated application has hardly been studied, it is still challenging to give a general conclusion on its effects on soil biota.

Table 4 summarizes the results of several studies showing changes in microbial communities following digestate application.

Table 4. Changes in the microbial community due to the application of digestates.

Digestate Raw Material	Dose	Soil	Test Plant	Change in Microbial Community	Reference
Sewage sludge	147 kg N ha ⁻¹	No data	Spring wheat	- Decline in diversity and richness, but effects on soil bacteria appears low - Digestate bacteria did not strongly alter community patterns in the soil - Digestate induced a shift in the soil microbial community	[162]
Dairy cattle manure + maize silage + agro-food waste	No data	Clay	No plant	- Digestate favoured some fungal groups like <i>Basidiomycota</i> - Increased abundance and number of bacterial species involved in N metabolism - Increase in the functional diversity of soil microbial communities	[166]
Sugar beet pulp + fruit marc + maize silage	100 g dw kg soil ⁻¹	Loam	Wheat	- Abundance enhancement of bacteria, actinobacteria and fungal communities - Increased the Gram+/Gram− ratio	[167]
Cow manure + silage	100 kg N ha ⁻¹	Sandy loam, clay loam	No plant	- Increased microbial biomass - Decreased microbial biomass C	[168]
Livestock and poultry manure + food waste	170 kg N ha ⁻¹	High and low nutrient soil	No plant	- Negative values of carbon use efficiency - Increased microbial biomass C	[169]
Livestock and poultry manure + food waste	170 kg N ha ⁻¹	High and low nutrient soil	No plant	- Increased fungal-to-bacterial ratio - Increased of carbon use efficiency - Increase in microbial abundance and diversity - Stabilization of the karyotic community	[169]
Municipal solid biowaste	Soil:digestate ratio was 14:1	Loamy sand	No plant	- The <i>Firmicutes</i> dominated community changed into a <i>Proteobacteria</i> and <i>Bacteroidota</i> dominated community - Increased the microbial biomass, not detailed community structure changes	[170]
Cattle slurry + maize silage	33 mg P kg soil ⁻¹	Loamy sand	Amaranth, Sorghum	- Increased the microbial biomass, not detailed community structure changes	[108]
Maize + grass + whole plant silage	33 mg P kg soil ⁻¹	Loamy sand	Amaranth, Sorghum	- Increased the microbial biomass, not detailed community structure changes - Negative effect on ammonia oxidizing archaea	[108]
Dairy cow manure	50 kg NH ₄ ⁺ ha ⁻¹	Andosol	No plant	- Positive effect on ammonia oxidizing bacteria if lime was also applied	[171]
Restaurant and agri-food waste + livestock effluents + plant matter	170 kg N ha ⁻¹	Loam	Maize, w. wheat, sugarbeet, spr. barley	- Increase in the abundance of the prokaryotic <i>Firmicutes</i> and the fungal <i>Mucoromycota</i> - No substantial alterations in the soil microbial structure - Increased number of genera - Increased relative abundance of <i>Myxococcota</i>	[172]
Agricultural and livestock waste	5 N unit 2 L soil ⁻¹	Clayey loam	No plant	- New genera appeared and involved in nutrient cycling and C storage - Microbial diversity remained stable - Increased the complexity of microbial community and bacterial interactions - Increase in Gram-negative bacteria	[173]
Pig slurry	340 kg N ha ⁻¹	Silty loam	No plant	- Decrease in fungi to bacteria ratio - Decreasing trend in both bacterial and yeast and fungal richness - Increase in AM fungal hyphae density at 50 kg N ha ⁻¹ dose - Shifted the bacterial community composition	[174]
Food waste	25 and 50 kg N ha ⁻¹	Yellow sandy duplex	Annual ryegrass	- Digestate affected bacterial, C and N cycling genes community composition	[97]

Table 4. Cont.

Digestate Raw Material	Dose	Soil	Test Plant	Change in Microbial Community	Reference
Cattle manure + clover/grass; chicken manure + cow and pig manure + ensiled energy crops	75 kg N ha ⁻¹	Silty clay loam	No plant	- Growth of saprotrophic soil fungi after solid and composted solid digestate - No effect of liquid fraction on microbial activity - Increased Shannon diversity and species richness of fungal communities, but not that of bacteria	[175]
No data	250–375 m ³ ha ⁻¹ yr ⁻¹	Sandy loam	Poplar	- Increased the relative abundance of <i>Gammaproteobacteria</i> and <i>Hyphomicrobiaceae</i> - More complex bacterial community and decrease of the complexity of fungal network - Increased the bacterial diversity and richness at digestate dose 165.1 t ha ⁻¹	[176]
Pig manure	150 kg N ha ⁻¹	Gleyic Solonchak, loam	Wheat, rice	- Decreased relative abundance of <i>Actinobacteria</i> - Increased relative abundance of <i>Nitrospirae</i> - Increased bacterial diversity	[85]
Pig manure	59.9–264.4 t ha ⁻¹	Silt loam	Rice and rape	- Increased the relative abundance of <i>Nitrospirae</i> - Decreased relative abundance of potential fungal pathogens (<i>Fusarium</i> , <i>Cylindrocarpon</i> , <i>Alternaria</i> , and <i>Phoma</i>) - Enriching in bacteria and increasing the phylogenetic relatedness of the bacterial community.	[177]
Pig manure	270–540 kg N ha ⁻¹	Mollic Endoaquept	Rice		[178]
Vegetable juice waste	0.6 g N kg soil ⁻¹	Latosols	No plant		[179]

6. Substances Limiting the Application of Digestates

In order to use digestates safely in the field, it is important to investigate their environmental risk potential. The range of pollutants that can be released into the environment with digestates and the degree of pollution risk are primarily determined by the biogas production process and the quality of the raw materials used, as well as by the soil and hydrogeographic properties of the given location. The legal regulations for the agricultural utilization of digestates from biogas production vary greatly from country to country. While there are countries where digestate exists as a legal category (e.g., Denmark), it is more common for it to be classified alongside other organic fertilizer materials, with no distinction made between them. Whether digestate is classified as a product or waste depends on its quality and the legal environment of the given country. Czatzkowska et al. [180] summarized the residues of heavy metals and antibiotics detected in digestates worldwide, mapped their risk, specified the permissible limits and pointed out the lack of legal regulation. Data on organic pollutants (TPH, PAH and BTEX) in digestates is particularly limited. Figure 4 shows the types of pollutants potentially entering the soil with digestates, with some typical examples, which will be presented in the following subsections.

6.1. Inorganic Pollutants

Some potentially toxic elements or heavy metals are required by organisms in small amounts and only become toxic above a certain concentration. This is true of anaerobic fermentation process, in which trace elements such as Fe, Ni, Co, Zn, Mo, Se, and W are required for the proper functioning of enzymes and coenzymes. However, excessive concentrations of these elements can restrict the agricultural use of digestates [181].

When assessing the risk of inorganic pollutants, the mobility of the given element in the soil and its availability to plants are very important aspects. In general, though their total element content does not change, the mobility and availability of potentially toxic elements

decrease during anaerobic digestion, primarily due to precipitation with sulfides, carbonates, and phosphates, but once they reach the soil, their biogenic relationships may change [26].



Figure 4. Pollutants potentially entering the soil with digestates.

The concentration of inorganic contaminants in digestates is often lower than in the waste streams of other organic fertilizers [182,183]. However, in some cases, the application of digestates may cause elements to accumulate in the soil. The application of pig manure-based digestate to arable land resulted in exceeding the limit values for Ni, Zn, Cd and Pb, and increasing the soil concentrations of Cu, Cr, As [184]. However, in other experiments, the concentration of toxic elements (Cu, Cd, Pb, Cr and Hg) did not exceed the relevant limit values in the soil or crops after several years of applying pig manure-based digestates [182,185]. In a field experiment, Wang et al. [186] demonstrated that the application of a diluted digestate resulted in significant Zn and Cu accumulation in the soil in a rice-growing area. In another Chinese long-term field experiment, Wang et al. [187] found that Zn accumulated slightly in the topsoil as a result of the digestate treatment, but remained below the safety limit, when pig manure-based digestate was applied at doses of 270 kg N ha⁻¹ and 540 kg N ha⁻¹.

In many respects, sewage sludge has similar effects on arable land as digestates, and although there is a risk of heavy metal contamination, only minimal enrichment of these pollutants was observed during the utilization of municipal sewage sludge [188,189].

6.2. Organic Pollutants

The minimal presence of organic contaminants does not affect the process of anaerobic digestion, but if digestion is incomplete and the organic contaminants are not degraded, they may accumulate in the digestates, reducing their usability. Such compounds include dioxins, PAHs, PCBs, chlorinated paraffins, phthalates and phenols [160]. Uncertainty may arise due to the wide range of pollutants for which regulation is scarce. These

include pharmaceutical residues, steroids, pollutants from hormone replacements, certain pesticides, herbicides and surfactants [190,191].

Biodegradable organic wastes contain drug residues such as carbamazepine, fluoxetine, triclosan, miconazole, ciprofloxacin, naproxen, tetracycline, doxycycline, clindamycin, erythromycin and clarithromycin. These compounds are very persistent and accumulate in the environment, and plants can also absorb them [192,193]. They proved to have negative effects on plant stem and root growth, and if they enter the food chain, they may cause passive drug consumption in both animals and humans [194]. Perfluorochemicals, polychlorinated alkanes and bisphenol compounds also deserve increased attention, as they affect the animal and human hormonal systems and are also carcinogenic [195].

The stability of drug residues in the soil is influenced by several factors. For example, the persistence of clotrimazole is reduced by high soil water content, while the opposite is true for fluconazole [196]. Soil pH and SOM content affect the degradation [197], and, together with the mineral fraction of the soil, the adsorption of drug residues [198]. If adsorbed on a soil constituent that is mobile, drug residues can easily reach the groundwater [199]. In Taiwan, the effect of a digestate made from animal waste on the presence of veterinary medicinal compounds in soil, plants, and water was investigated using a library containing 1068 compounds. Antibiotics were not detected in plants, but they were detected in soil and digestates [200]. Widyasari-Mehta et al. [201] identified 34 different antibiotics in pig-slurry-based digestates in Germany, of which tetracyclines caused the most concern. Lehmann and Bloem [202] examined antibiotic contamination in 29 biogas plants in Sweden, Finland, and Germany and found that anaerobic digestion did not reduce the presence of antibiotics in the digestate, so these contaminants might enter the soil when digestates are applied.

In addition to pharmaceutical residues, other organic contaminants can also limit the usability of digestates. Ali et al. [203] collected 19 biogas digestate samples from 12 different biogas plants in Norway to examine organic contaminants of emerging concern, and found that even the optimized anaerobic digestion process did not effectively remove organic micropollutants associated with the substrates. The most worrying compounds were the sunscreen octocrylene, acetaminophen (paracetamol) and the flame retardant TCPP, which were found in high concentrations.

Hormone-like compounds may also appear in digestates. Rodriguez-Navas et al. [204] examined hormone compounds (estrogens, androgens, progesterones) in the sludge remaining after biogas production in two biogas plants using different digestion technologies. The total hormone concentration in a digestate obtained from a 95:5 ratio of pig manure: distillery waste was 11.2 mg kg^{-1} , while in a digestate obtained from a 75–80:10–20:5–10 ratio of pig manure: slaughterhouse waste: food industry waste it was 52.5 mg kg^{-1} . Based on these results, the technologies applied were unable to eliminate the hormones accumulated in the sludge.

6.3. Microplastics

Microplastics, which are present in soil, air and water, are also a potential pollutant in the digestates [205]. The size of microplastics ranges from $1 \text{ }\mu\text{m}$ to 1 mm . The original sources of microplastics are polyethylene, polypropylene, polyvinyl chloride, polyethylene terephthalate and other polymers [206]. Since microplastics are non-biodegradable but sensitive to photodegradation, they can persist in the environment for decades [207]. There is extensive literature on the effects, primarily in aquatic ecosystems, where they are known to have negative effects on DNA, biodiversity [208], reproduction and the function of certain enzymes [209].

Microplastics also affect the anaerobic digestion process by reducing the activity of microorganisms [210–212]. Some studies found that anaerobic digestion reduced the amount of microplastics in digestate [213], while others reported no effect [214].

During wastewater treatment, a significant proportion of microplastics, which can also bind heavy metals to their surface, end up in the sludge [213,215] and the soil application of various organic materials, such as digestate, to arable land can be a primary source of microplastic accumulation in soils. This may affect not only terrestrial ecosystems but also freshwater and marine ecosystems, while humans are also exposed to these pollutants [216]. When microplastics enter the soil, they affect its water, heat and air balance, soil chemistry, nutrient management and biological processes [217,218].

Although reports on targeted research on the effect of the digestates remaining during biogas production on the microplastic content of soils are scarce, studies on sewage sludge and microplastics suggest that microplastics may also be present in digestates if the raw material is, for example, sewage sludge or certain agricultural waste. Therefore, their accumulation in the soil is to be expected.

6.4. Microbial Pollutants

Many microbial contaminants are pathogens. Some authors found that anaerobic digestion drastically reduced the number of these pathogens [219], but other studies demonstrated that they not only survived the digestion process, but also reproduced when introduced into the soil [220].

Bacteria such as *Escherichia coli* or *Salmonella* spp. could not be cultured from the digestate after 60 days of anaerobic fermentation at 37 °C [221], in contrast to *Listeria* spp. A similar result was obtained by Bonetta et al. [222]. Islam et al. [223] investigated the potential for pathogens to be cultured from the materials of the anaerobic digestion process in Bangladesh. Although *Escherichia coli*, *Salmonella* spp., and *Staphylococcus* spp. were present in the raw materials, they were no longer cultured from the sludge after 60 days of anaerobic digestion. Nag et al. [224] developed an assessment method to rank the potential health risk of pathogens from digestate based on various parameters. Using multiple sources and evaluating different feedstocks and digestion technologies, they concluded that *Cryptosporidium parvum*, *Salmonella* spp., *norovirus*, *Streptococcus pyogenes*, *enteropathogenic Escherichia coli*, *Mycobacterium* spp., *Salmonella typhi*, *Salmonella paratyphi*, *Clostridium* spp., *Listeria monocytogenes* and *Campylobacter coli* posed the highest human health risk. According to Massé et al. [225], in addition to the above, the following pathogens can also be detected in materials intended for anaerobic digestion: *Yersinia enterocolitica*, *Staphylococcus* spp. and *Streptococcus* spp.

In connection with the use of digestates, testing for antibiotic-resistant bacteria in the environment and especially in the soil-plant system, may be justified. According to the results of Pu et al. [226], the number of bacteria resistant to the tetracycline and MLSB antibiotic groups decreased significantly in digestate made from sewage sludge and pig manure during the anaerobic digestion process (reactor temperature 26–36 °C), but the number of bacteria resistant to certain sulfonamides, aminoglycosides, florfenicol, chloramphenicol, and amphenicol multiplied. Lu et al. [227] found in a field experiment that the long-term application of digestate (8–18 years) increased the concentration of tetracyclines in the soil, while also greatly increasing the number of antibiotic-resistant bacteria and the transposase gene that promotes antibiotic resistance.

7. Conclusions

In the future, the amount of digestates, a by-product of anaerobic digestion, can be expected to increase, as well as its use in agriculture. In addition to increasing the gas yield

of anaerobic digestion, more attention should be paid to the properties of the resulting digestate to ensure that it can be safely applied to soil without endangering its health or that of the environment. Digestates predominantly have a slightly alkaline pH, and a significant content of macro-, and microelements and OM. The properties of the digestates produced by biogas plants are greatly affected by the raw materials used and the technological parameters employed. This means that digestates have different properties, so they affect the soil in different ways. The diversity of soil properties, such as texture, pH, CEC and OM content is also an important factor influencing the effect of digestates.

Digestates fundamentally affect soil chemistry by reducing acidity through their high pH and by introducing stable organic matter into the soil, thereby improving its structure. Digestates affect the water holding capacity, wettability, infiltration indicators, dispersibility, bulk density, porosity, aggregate stability, and the amount of aggregates. In general, the bulk density of the soil decreases and its water holding capacity and the proportion of water-resistant aggregates increase, but the trends are not always clearly positive. Digestates are rich in plant nutrients, especially in plant available $\text{NH}_4\text{-N}$ but can also increase the available P, K and micronutrients in the soil. To determine the application rates, it is necessary to examine the N content of the digestate to avoid excessive $\text{NH}_4\text{-N}$ application. The long-term and large-scale application of certain high-salt digestates, typically made from food waste, can pose the potential risk of soil salinization. Appropriate dilution or irrigation should be provided when applying such digestates. Initially, digestates stimulate soil microbial activity and enzyme functions through their readily degradable organic carbon. However, this effect may diminish in the long term due to the depletion of readily metabolizable carbon. Therefore, repeated applications over a long period of time may be necessary to maintain this effect. Although digestates can improve the physical, chemical and biological properties of the soil, but they may also contain harmful components that limit their application to certain soils or areas. The raw materials and processing method fundamentally determine the concentration of contaminants in digestates. Inorganic, organic and microbiological pollutants, as well as microplastics may occur in digestates. The presence of antibiotic-resistant microorganisms is noteworthy. It is recommended that biogas production processes should be optimized to reduce pollutant emissions and that waste should be separated at the site of its generation, and should be collected selectively, especially in the case of municipal solid waste and food waste. For safe application, it is necessary to examine both the digestates and the soil conditions of the area of use. Not only the nutrient content, primarily N of the digestate, but also its pH, salt content, organic and microbiological components, as well as pollutant content must be taken into account when determining optimal doses, because the relevant soil properties could be significantly influenced. In the case of longer-term application, regular soil analysis is recommended for these parameters.

Caution should be exercised when interpreting and generalizing the results of digestate-related research. The greatest limitation to interpreting the results is probably the diversity of digestate compositions. As the various raw materials, digestion processes and effects on soil properties were presented, along with the limiting substances, it became evident how diverse the properties and quality parameters of digestates can be. Another major factor limiting the reliability of experimental conclusions is the diversity of soil types. Additional limitations to the applicability of experimental results may include different climatic factors encountered in field experiments, the difficulty of applying laboratory results to field conditions with complex environmental interactions, and the duration of the experiments. Studies examining short-term effects may not reflect the effects of long-term or repeated digestate application to soil.

8. Future Directions

The results indicate that the application of digestates has mostly positive effects, so their use as fertilizer and soil amendments in agriculture is promising. However, significantly less research has been reported on the agricultural utilization of digestates than, for example, on that of sewage sludge. There is urgent need to test microplastics in connection with the application of digestates. Much more extensive investigation of antibiotic-resistant microorganisms released into the environment with digestates should be made in the near future. Further, detailed research involving different climatic conditions, soil types and test plants is needed to find out how digestates of different compositions, made from different raw materials in the biogas production process can be safely applied to the soil. This research should also investigate how they affect the physical, chemical and OM transformation processes of the soil and what environmental risks are posed by their long-term, regular application.

Author Contributions: Conceptualization: M.R. and P.R.; writing—original draft preparation of Section 1: M.R.; Section 2: P.R.; Section 3: O.S. and M.R.; Section 4: M.R.; Section 5: M.R., O.S., P.R., M.M. and A.S.; Section 6: N.U.; writing—review and editing: P.R., O.S. and M.R.; All authors have read and agreed to the published version of the manuscript.

Funding: The work was funded by the Hungarian Ministry of Agriculture under Agreement No. AGMF/54/2024(3) entitled “Biogas digestate as fertilizer and soil amendment”.

Data Availability Statement: No new data were created in this study.

Conflicts of Interest: The authors declare no conflicts of interest. The funders had no role in the design of the study; in the collection, analyses, or interpretation of data; in the writing of the manuscript; or in the decision to publish the results.

Abbreviations

The following abbreviations are used in this manuscript:

CEC	cation exchange capacity
EC	electrical conductivity
MSW	municipal solid waste
OC	organic carbon
OM	organic matter
SOM	soil organic matter

References

1. Maalouf, A.; Mavropoulos, A. Re-Assessing Global Municipal Solid Waste Generation. *Waste Manag. Res.* **2023**, *41*, 936–947. [CrossRef] [PubMed]
2. United Nations Environment Programme (UNEP). *Global Waste Management Outlook 2024: Beyond an Age of Waste—Turning Rubbish into a Resource*; UNEP: Nairobi, Kenya, 2024; ISBN 978-92-807-4129-2. [CrossRef]
3. Eurostat Municipal Waste Statistics. Available online: https://ec.europa.eu/eurostat/statistics-explained/index.php?title=Municipal_waste_statistics (accessed on 18 June 2025).
4. Günther, S.; Karras, T.; Naegeli de Torres, F.; Semella, S.; Thrän, D. Temporal and Spatial Mapping of Theoretical Biomass Potential across the European Union. *Earth Syst. Sci. Data* **2024**, *16*, 59–74. [CrossRef]
5. Dou, Z.; Toth, J.D. Global Primary Data on Consumer Food Waste: Rate and Characteristics—A Review. *Resour. Conserv. Recycl.* **2021**, *168*, 105332. [CrossRef]
6. Elgarahy, A.M.; Eloffy, M.G.; Priya, A.K.; Yogeshwaran, V.; Yang, Z.; Elwakeel, K.Z.; Lopez-Maldonado, E.A. Biosolids Management and Utilizations: A Review. *J. Clean. Prod.* **2024**, *451*, 141974. [CrossRef]
7. Czekala, W.; Nowak, M.; Bojarski, W. Anaerobic Digestion and Composting as Methods of Bio-Waste Management. *Agric. Eng.* **2023**, *27*, 173–186. [CrossRef]

8. Bhunia, S.; Bhowmik, A.; Mallick, R.; Mukherjee, J. Agronomic Efficiency of Animal-Derived Organic Fertilizers and Their Effects on Biology and Fertility of Soil: A Review. *Agronomy* **2021**, *11*, 823. [\[CrossRef\]](#)
9. Montusiewicz, A. *Methods for Enhancing Biogas Production*, 1st ed.; Routledge: London, UK, 2024; ISBN 978-1-003-00354-0.
10. Zupančič, M.; Možic, V.; Može, M.; Cimerman, F.; Golobič, I. Current Status and Review of Waste-to-Biogas Conversion for Selected European Countries and Worldwide. *Sustainability* **2022**, *14*, 1823. [\[CrossRef\]](#)
11. Jurgutis, L.; Šlepetienė, A.; Amalevičiūtė-Volungė, K.; Volungevičius, J.; Šlepetys, J. The Effect of Digestate Fertilisation on Grass Biogas Yield and Soil Properties in Field-Biomass-Biogas-Field Renewable Energy Production Approach in Lithuania. *Biomass Bioenergy* **2021**, *153*, 106211. [\[CrossRef\]](#)
12. Soha, T.; Papp, L.; Csontos, C.; Munkácsy, B. The Importance of High Crop Residue Demand on Biogas Plant Site Selection, Scaling and Feedstock Allocation—A Regional Scale Concept in a Hungarian Study Area. *Renew. Sustain. Energy Rev.* **2021**, *141*, 110822. [\[CrossRef\]](#)
13. Sobhi, M.; Guo, J.; Gaballah, M.S.; Li, B.; Zheng, J.; Cui, X.; Sun, H.; Dong, R. Selecting the Optimal Nutrients Recovery Application for a Biogas Slurry Based on Its Characteristics and the Local Environmental Conditions: A Critical Review. *Sci. Total Environ.* **2022**, *814*, 152700. [\[CrossRef\]](#)
14. Subbarao, P.M.V.; D' Silva, T.C.; Adlak, K.; Kumar, S.; Chandra, R.; Vijay, V.K. Anaerobic Digestion as a Sustainable Technology for Efficiently Utilizing Biomass in the Context of Carbon Neutrality and Circular Economy. *Environ. Res.* **2023**, *234*, 116286. [\[CrossRef\]](#) [\[PubMed\]](#)
15. Czekala, W.; Lewicki, A.; Pochwatka, P.; Czekala, A.; Wojcieszak, D.; Jóźwiakowski, K.; Waliszewska, H. Digestate Management in Polish Farms as an Element of the Nutrient Cycle. *J. Clean. Prod.* **2020**, *242*, 118454. [\[CrossRef\]](#)
16. Ellacuriaga, M.; García-Cascallana, J.; Gómez, X. Biogas Production from Organic Wastes: Integrating Concepts of Circular Economy. *Fuels* **2021**, *2*, 144–167. [\[CrossRef\]](#)
17. Logan, M.; Visvanathan, C. Management Strategies for Anaerobic Digestate of Organic Fraction of Municipal Solid Waste: Current Status and Future Prospects. *Waste Manag. Res. J. Sustain. Circ. Econ.* **2019**, *37*, 27–39. [\[CrossRef\]](#)
18. Mukhtiar, A.; Mahmood, A.; Zia, M.A.; Ameen, M.; Dong, R.; Shoujun, Y.; Javaid, M.M.; Khan, B.A.; Nadeem, M.A. Role of Biogas Slurry to Reclaim Soil Properties Providing an Eco-Friendly Approach for Crop Productivity. *Bioresour. Technol. Rep.* **2024**, *25*, 101716. [\[CrossRef\]](#)
19. Kovačić, Đ.; Lončarić, Z.; Jović, J.; Samac, D.; Popović, B.; Tišma, M. Digestate Management and Processing Practices: A Review. *Appl. Sci.* **2022**, *12*, 9216. [\[CrossRef\]](#)
20. Chojnacka, K.; Mikula, K.; Skrzypczak, D.; Izydorczyk, G.; Gorazda, K.; Kulczycka, J.; Kominko, H.; Moustakas, K.; Witek-Krowiak, A. Practical Aspects of Biowastes Conversion to Fertilizers. *Biomass Convers. Biorefinery* **2024**, *14*, 1515–1533. [\[CrossRef\]](#)
21. Guilayn, F.; Jimenez, J.; Martel, J.-L.; Rouez, M.; Crest, M.; Patureau, D. First Fertilizing-Value Typology of Digestates: A Decision-Making Tool for Regulation. *Waste Manag.* **2019**, *86*, 67–79. [\[CrossRef\]](#)
22. Chozhavendhan, S.; Karthigadevi, G.; Bharathiraja, B.; Praveen Kumar, R.; Abo, L.D.; Venkatesa Prabhu, S.; Balachandar, R.; Jayakumar, M. Current and Prognostic Overview on the Strategic Exploitation of Anaerobic Digestion and Digestate: A Review. *Environ. Res.* **2023**, *216*, 114526. [\[CrossRef\]](#)
23. Karimi, B.; Sadet-Bourgeteau, S.; Cannavacciuolo, M.; Chauvin, C.; Flamin, C.; Haumont, A.; Jean-Baptiste, V.; Reibel, A.; Vrignaud, G.; Ranjard, L. Impact of Biogas Digestates on Soil Microbiota in Agriculture: A Review. *Environ. Chem. Lett.* **2022**, *20*, 3265–3288. [\[CrossRef\]](#)
24. van Midden, C.; Harris, J.; Shaw, L.; Sizmur, T.; Pawlett, M. The Impact of Anaerobic Digestate on Soil Life: A Review. *Appl. Soil Ecol.* **2023**, *191*, 105066. [\[CrossRef\]](#)
25. O'Connor, J.; Mickan, B.S.; Rinklebe, J.; Song, H.; Siddique, K.H.M.; Wang, H.; Kirkham, M.B.; Bolan, N.S. Environmental Implications, Potential Value, and Future of Food-Waste Anaerobic Digestate Management: A Review. *J. Environ. Manag.* **2022**, *318*, 115519. [\[CrossRef\]](#) [\[PubMed\]](#)
26. Möller, K.; Müller, T. Effects of Anaerobic Digestion on Digestate Nutrient Availability and Crop Growth: A Review. *Eng. Life Sci.* **2012**, *12*, 242–257. [\[CrossRef\]](#)
27. Nkoa, R. Agricultural Benefits and Environmental Risks of Soil Fertilization with Anaerobic Digestates: A Review. *Agron. Sustain. Dev.* **2014**, *34*, 473–492. [\[CrossRef\]](#)
28. Bünemann, E.K.; Bongiorno, G.; Bai, Z.; Creamer, R.E.; De Deyn, G.; de Goede, R.; Fleskens, L.; Geissen, V.; Kuyper, T.W.; Mäder, P.; et al. Soil Quality—A Critical Review. *Soil Biol. Biochem.* **2018**, *120*, 105–125. [\[CrossRef\]](#)
29. Shi, L.; Simplicio, W.S.; Wu, G.; Hu, Z.; Hu, H.; Zhan, X. Nutrient Recovery from Digestate of Anaerobic Digestion of Livestock Manure: A Review. *Curr. Pollut. Rep.* **2018**, *4*, 74–83. [\[CrossRef\]](#)
30. Uddin, M.M.; Wright, M.M. Anaerobic Digestion Fundamentals, Challenges, and Technological Advances. *Phys. Sci. Rev.* **2023**, *8*, 2819–2837. [\[CrossRef\]](#)
31. Latif, M.A.; Mehta, C.M.; Batstone, D.J. Influence of Low pH on Continuous Anaerobic Digestion of Waste Activated Sludge. *Water Res.* **2017**, *113*, 42–49. [\[CrossRef\]](#)

32. Kovács, A.B.; Kremper, R.; Kincses, I.; Leviczky, Á. Influences of Different Organic Fertilizers on Nutrients of Humic Sandy Soil and on the Growth of Spinach (*Spinacia oleracea* L.). *Acta Agrar. Debreceniensis* **2016**, *70*, 23–28. [\[CrossRef\]](#)
33. Iocoli, G.A.; Zabaloy, M.C.; Pasdevicelli, G.; Gómez, M.A. Use of Biogas Digestates Obtained by Anaerobic Digestion and Co-Digestion as Fertilizers: Characterization, Soil Biological Activity and Growth Dynamic of *Lactuca sativa* L. *Sci. Total Environ.* **2019**, *647*, 11–19. [\[CrossRef\]](#)
34. Pampillón-González, L.; Luna-Guido, M.; Ruíz-Valdiviezo, V.M.; Franco-Hernández, O.; Fernández-Luqueño, F.; Paredes-López, O.; Hernández, G.; Dendooven, L. Greenhouse Gas Emissions and Growth of Wheat Cultivated in Soil Amended with Digestate from Biogas Production. *Pedosphere* **2017**, *27*, 318–327. [\[CrossRef\]](#)
35. Mendonça Costa, M.S.S.D.; Lucas, J.D.; Mendonça Costa, L.A.D.; Orrico, A.C.A. A Highly Concentrated Diet Increases Biogas Production and the Agronomic Value of Young Bull's Manure. *Waste Manag.* **2016**, *48*, 521–527. [\[CrossRef\]](#) [\[PubMed\]](#)
36. Alfa, M.I.; Adie, D.B.; Igboro, S.B.; Oranusi, U.S.; Dahunsi, S.O.; Akali, D.M. Assessment of Biofertilizer Quality and Health Implications of Anaerobic Digestion Effluent of Cow Dung and Chicken Droppings. *Renew. Energy* **2014**, *63*, 681–686. [\[CrossRef\]](#)
37. Nölvak, H.; Truu, M.; Kanger, K.; Tampere, M.; Espenberg, M.; Loit, E.; Raave, H.; Truu, J. Inorganic and Organic Fertilizers Impact the Abundance and Proportion of Antibiotic Resistance and Integron-Integrase Genes in Agricultural Grassland Soil. *Sci. Total Environ.* **2016**, *562*, 678–689. [\[CrossRef\]](#) [\[PubMed\]](#)
38. Lamolinara, B.; Pérez-Martínez, A.; Guardado-Yordi, E.; Guillén Fiallos, C.; Diéguez-Santana, K.; Ruiz-Mercado, G.J. Anaerobic Digestate Management, Environmental Impacts, and Techno-Economic Challenges. *Waste Manag.* **2022**, *140*, 14–30. [\[CrossRef\]](#)
39. Momayez, F.; Karimi, K.; Taherzadeh, M.J. Energy Recovery from Industrial Crop Wastes by Dry Anaerobic Digestion: A Review. *Ind. Crops Prod.* **2019**, *129*, 673–687. [\[CrossRef\]](#)
40. Zhao, J.; Liu, Y.; Wang, D.; Chen, F.; Li, X.; Zeng, G.; Yang, Q. Potential Impact of Salinity on Methane Production from Food Waste Anaerobic Digestion. *Waste Manag.* **2017**, *67*, 308–314. [\[CrossRef\]](#)
41. Slepeliene, A.; Ceseviciene, J.; Amaleviciute-Volunge, K.; Mankeviciene, A.; Parasotas, I.; Skersiene, A.; Jurgutis, L.; Volungevicius, J.; Veteikis, D.; Mockeviciene, I. Solid and Liquid Phases of Anaerobic Digestate for Sustainable Use of Agricultural Soil. *Sustainability* **2023**, *15*, 1345. [\[CrossRef\]](#)
42. Ren, Y.; Yu, M.; Wu, C.; Wang, Q.; Gao, M.; Huang, Q.; Liu, Y. A Comprehensive Review on Food Waste Anaerobic Digestion: Research Updates and Tendencies. *Bioresour. Technol.* **2018**, *247*, 1069–1076. [\[CrossRef\]](#)
43. Nicholson, F.; Bhogal, A.; Cardenas, L.; Chadwick, D.; Misselbrook, T.; Rollett, A.; Taylor, M.; Thorman, R.; Williams, J. Nitrogen Losses to the Environment Following Food-Based Digestate and Compost Applications to Agricultural Land. *Environ. Pollut.* **2017**, *228*, 504–516. [\[CrossRef\]](#)
44. Lu, J.; Xu, S. Post-Treatment of Food Waste Digestate towards Land Application: A Review. *J. Clean. Prod.* **2021**, *303*, 127033. [\[CrossRef\]](#)
45. O'Connor, J.; Mickan, B.S.; Siddique, K.H.M.; Rinklebe, J.; Kirkham, M.B.; Bolan, N.S. Physical, Chemical, and Microbial Contaminants in Food Waste Management for Soil Application: A Review. *Environ. Pollut.* **2022**, *300*, 118860. [\[CrossRef\]](#) [\[PubMed\]](#)
46. Pawlett, M.; Tibbett, M. Is Sodium in Anaerobically Digested Food Waste a Potential Risk to Soils? *Sustain. Environ. Res.* **2015**, *25*, 235–239.
47. Cheong, J.C.; Lee, J.T.E.; Lim, J.W.; Song, S.; Tan, J.K.N.; Chiam, Z.Y.; Yap, K.Y.; Lim, E.Y.; Zhang, J.; Tan, H.T.W.; et al. Closing the Food Waste Loop: Food Waste Anaerobic Digestate as Fertilizer for the Cultivation of the Leafy Vegetable, Xiao Bai Cai (*Brassica rapa*). *Sci. Total Environ.* **2020**, *715*, 136789. [\[CrossRef\]](#)
48. Campuzano, R.; González-Martínez, S. Characteristics of the Organic Fraction of Municipal Solid Waste and Methane Production: A Review. *Waste Manag.* **2016**, *54*, 3–12. [\[CrossRef\]](#)
49. Tyagi, V.K.; Fdez-Güelfo, L.A.; Zhou, Y.; Álvarez-Gallego, C.J.; Garcia, L.I.R.; Ng, W.J. Anaerobic Co-Digestion of Organic Fraction of Municipal Solid Waste (OFMSW): Progress and Challenges. *Renew. Sustain. Energy Rev.* **2018**, *93*, 380–399. [\[CrossRef\]](#)
50. Patziger, M.; Kiss, K. Towards a Hydrodynamically Enhanced Design and Operation of Primary Settling Tanks—Results of a Long Term *in Situ* Measurement Investigation Program. *Water Environ. J.* **2015**, *29*, 338–345. [\[CrossRef\]](#)
51. Cao, Y.; Pawłowski, A. Sewage Sludge-to-Energy Approaches Based on Anaerobic Digestion and Pyrolysis: Brief Overview and Energy Efficiency Assessment. *Renew. Sustain. Energy Rev.* **2012**, *16*, 1657–1665. [\[CrossRef\]](#)
52. Dandikas, V.; Heuwinkel, H.; Lichti, F.; Drewes, J.E.; Koch, K. Correlation between Biogas Yield and Chemical Composition of Energy Crops. *Bioresour. Technol.* **2014**, *174*, 316–320. [\[CrossRef\]](#)
53. Jaffar, M.; Pang, Y.; Yuan, H.; Zou, D.; Liu, Y.; Zhu, B.; Korai, R.M.; Li, X. Wheat Straw Pretreatment with KOH for Enhancing Biomethane Production and Fertilizer Value in Anaerobic Digestion. *Chin. J. Chem. Eng.* **2016**, *24*, 404–409. [\[CrossRef\]](#)
54. Provenzano, M.R.; Cavallo, O.; Malerba, A.D.; Fabbri, C.; Zacccone, C. Unravelling (Maize silage) Digestate Features throughout a Full-Scale Plant: A Spectroscopic and Thermal Approach. *J. Clean. Prod.* **2018**, *193*, 372–378. [\[CrossRef\]](#)
55. Czako-Vér, K.; Somfai, D.; Suhajda, E.; Strasszer, M.; Árvay, G.; Dolgosné Kovács, A.; Dittrich, E. The Application of Biogas Fermentation Digestate as Soil Fertilizer. *Analecta Tech. Szeged.* **2020**, *14*, 76–81. [\[CrossRef\]](#)

56. Negri, M.; Bacenetti, J.; Fiala, M.; Bocchi, S. Evaluation of Anaerobic Degradation, Biogas and Digestate Production of Cereal Silages Using Nylon-Bags. *Bioresour. Technol.* **2016**, *209*, 40–49. [[CrossRef](#)] [[PubMed](#)]
57. Luste, S.; Heinonen-Tanski, H.; Luostarinen, S. Co-Digestion of Dairy Cattle Slurry and Industrial Meat-Processing by-Products—Effect of Ultrasound and Hygienization Pre-Treatments. *Bioresour. Technol.* **2012**, *104*, 195–201. [[CrossRef](#)]
58. Sánchez-Muñoz, S.; Barbosa, F.G.; Jiménez-Ascencio, J.; Mier-Alba, E.; Singh, A.K.; Dos Santos, J.C.; Balagurusamy, N.; Da Silva, S.S.; Chandel, A.K. Technological Routes for Biogas Production: Current Status and Future Perspectives. In *Biogas Production*; Balagurusamy, N., Chandel, A.K., Eds.; Springer International Publishing: Cham, Switzerland, 2020; pp. 3–17. ISBN 978-3-030-58826-7.
59. Qian, S.; Chen, L.; Xu, S.; Zeng, C.; Lian, X.; Xia, Z.; Zou, J. Research on Methane-Rich Biogas Production Technology by Anaerobic Digestion Under Carbon Neutrality: A Review. *Sustainability* **2025**, *17*, 1425. [[CrossRef](#)]
60. Zhou, J.; Zhang, Y.; Khoshnevisan, B.; Duan, N. Meta-Analysis of Anaerobic Co-Digestion of Livestock Manure in Last Decade: Identification of Synergistic Effect and Optimization Synergy Range. *Appl. Energy* **2021**, *282*, 116128. [[CrossRef](#)]
61. Karki, R.; Chuenchart, W.; Surendra, K.C.; Shrestha, S.; Raskin, L.; Sung, S.; Hashimoto, A.; Kumar Khanal, S. Anaerobic Co-Digestion: Current Status and Perspectives. *Bioresour. Technol.* **2021**, *330*, 125001. [[CrossRef](#)]
62. Grigatti, M. The Impact of Raw and Composted Food Waste Anaerobic Digestates on Soil Organic Carbon Management: A Pot Study. *Waste Biomass Valorization* **2024**, *15*, 4915–4925. [[CrossRef](#)]
63. Rigby, H.; Smith, S.R. Nitrogen Availability and Indirect Measurements of Greenhouse Gas Emissions from Aerobic and Anaerobic Biowaste Digestates Applied to Agricultural Soils. *Waste Manag.* **2013**, *33*, 2641–2652. [[CrossRef](#)]
64. Ronga, D.; Setti, L.; Salvarani, C.; De Leo, R.; Bedin, E.; Pulvirenti, A.; Milc, J.; Pecchioni, N.; Francia, E. Effects of Solid and Liquid Digestate for Hydroponic Baby Leaf Lettuce (*Lactuca sativa* L.) Cultivation. *Sci. Hortic.* **2019**, *244*, 172–181. [[CrossRef](#)]
65. Koch, K.; Plabst, M.; Schmidt, A.; Helmreich, B.; Drewes, J.E. Co-Digestion of Food Waste in a Municipal Wastewater Treatment Plant: Comparison of Batch Tests and Full-Scale Experiences. *Waste Manag.* **2016**, *47*, 28–33. [[CrossRef](#)] [[PubMed](#)]
66. Li, R.; Duan, N.; Zhang, Y.; Liu, Z.; Li, B.; Zhang, D.; Lu, H.; Dong, T. Co-Digestion of Chicken Manure and Microalgae *Chlorella* 1067 Grown in the Recycled Digestate: Nutrients Reuse and Biogas Enhancement. *Waste Manag.* **2017**, *70*, 247–254. [[CrossRef](#)] [[PubMed](#)]
67. Drózd, D.; Wystalska, K.; Malińska, K.; Grosser, A.; Grobelak, A.; Kacprzak, M. Management of Poultry Manure in Poland—Current State and Future Perspectives. *J. Environ. Manag.* **2020**, *264*, 110327. [[CrossRef](#)] [[PubMed](#)]
68. Neshat, S.A.; Mohammadi, M.; Najafpour, G.D.; Lahijani, P. Anaerobic Co-Digestion of Animal Manures and Lignocellulosic Residues as a Potent Approach for Sustainable Biogas Production. *Renew. Sustain. Energy Rev.* **2017**, *79*, 308–322. [[CrossRef](#)]
69. Muscolo, A.; Settineri, G.; Papalia, T.; Attinà, E.; Basile, C.; Panuccio, M.R. Anaerobic Co-Digestion of Recalcitrant Agricultural Wastes: Characterizing of Biochemical Parameters of Digestate and Its Impacts on Soil Ecosystem. *Sci. Total Environ.* **2017**, *586*, 746–752. [[CrossRef](#)]
70. Patel, V.; Pandit, S.; Chandrasekhar, K. Basics of Methanogenesis in Anaerobic Digester. In *Microbial Applications Vol. 2*; Kalia, V.C., Ed.; Springer International Publishing: Cham, Switzerland, 2017; pp. 291–314. ISBN 978-3-319-52668-3.
71. Abubaker, J.; Risberg, K.; Pell, M. Biogas Residues as Fertilisers—Effects on Wheat Growth and Soil Microbial Activities. *Appl. Energy* **2012**, *99*, 126–134. [[CrossRef](#)]
72. Nowak, M.; Czekala, W. Sustainable Use of Digestate from Biogas Plants: Separation of Raw Digestate and Liquid Fraction Processing. *Sustainability* **2024**, *16*, 5461. [[CrossRef](#)]
73. Schievano, A.; D’Imporzano, G.; Salati, S.; Adani, F. On-Field Study of Anaerobic Digestion Full-Scale Plants (Part I): An on-Field Methodology to Determine Mass, Carbon and Nutrients Balance. *Bioresour. Technol.* **2011**, *102*, 7737–7744. [[CrossRef](#)]
74. Kim, J.; Lee, C. Response of a Continuous Anaerobic Digester to Temperature Transitions: A Critical Range for Restructuring the Microbial Community Structure and Function. *Water Res.* **2016**, *89*, 241–251. [[CrossRef](#)]
75. Meegoda, J.N.; Li, B.; Patel, K.; Wang, L.B. A Review of the Processes, Parameters, and Optimization of Anaerobic Digestion. *Int. J. Environ. Res. Public Health* **2018**, *15*, 2224. [[CrossRef](#)]
76. Wang, Z.; Hu, Y.; Wang, S.; Wu, G.; Zhan, X. A Critical Review on Dry Anaerobic Digestion of Organic Waste: Characteristics, Operational Conditions, and Improvement Strategies. *Renew. Sustain. Energy Rev.* **2023**, *176*, 113208. [[CrossRef](#)]
77. Menardo, S.; Gioelli, F.; Balsari, P. The Methane Yield of Digestate: Effect of Organic Loading Rate, Hydraulic Retention Time, and Plant Feeding. *Bioresour. Technol.* **2011**, *102*, 2348–2351. [[CrossRef](#)] [[PubMed](#)]
78. Latif, M.A.; Mehta, C.M.; Batstone, D.J. Low pH Anaerobic Digestion of Waste Activated Sludge for Enhanced Phosphorous Release. *Water Res.* **2015**, *81*, 288–293. [[CrossRef](#)] [[PubMed](#)]
79. De Neve, S.; Sleutel, S.; Hofman, G. Carbon Mineralization from Composts and Food Industry Wastes Added to Soil. *Nutr. Cycl. Agroecosyst.* **2003**, *67*, 13–20. [[CrossRef](#)]
80. Glowacka, A.; Szostak, B.; Klebaniuk, R. Effect of Biogas Digestate and Mineral Fertilisation on the Soil Properties and Yield and Nutritional Value of Switchgrass Forage. *Agronomy* **2020**, *10*, 490. [[CrossRef](#)]

81. Smith, J.; Abegaz, A.; Matthews, R.B.; Subedi, M.; Orskov, E.R.; Tumwesige, V.; Smith, P. What Is the Potential for Biogas Digesters to Improve Soil Carbon Sequestration in Sub-Saharan Africa? Comparison with Other Uses of Organic Residues. *Biomass Bioenergy* **2014**, *70*, 73–86. [\[CrossRef\]](#)
82. Tambone, F.; Genevini, P.; D’Imporzano, G.; Adani, F. Assessing Amendment Properties of Digestate by Studying the Organic Matter Composition and the Degree of Biological Stability during the Anaerobic Digestion of the Organic Fraction of MSW. *Bioresour. Technol.* **2009**, *100*, 3140–3142. [\[CrossRef\]](#)
83. Li, H.; Li, Y.; Li, C. Evolution of humic substances during anaerobic sludge digestion. *Environ. Eng. Manag. J.* **2017**, *16*, 1577–1582. [\[CrossRef\]](#)
84. Slepeliene, A.; Kochiieru, M.; Jurgutis, L.; Mankeviciene, A.; Skersiene, A.; Belova, O. The Effect of Anaerobic Digestate on the Soil Organic Carbon and Humified Carbon Fractions in Different Land-Use Systems in Lithuania. *Land* **2022**, *11*, 133. [\[CrossRef\]](#)
85. Tang, Y.; Luo, L.; Carswell, A.; Misselbrook, T.; Shen, J.; Han, J. Changes in Soil Organic Carbon Status and Microbial Community Structure Following Biogas Slurry Application in a Wheat-Rice Rotation. *Sci. Total Environ.* **2021**, *757*, 143786. [\[CrossRef\]](#)
86. Li, J.; Li, H.; Zhang, Q.; Shao, H.; Gao, C.; Zhang, X. Effects of Fertilization and Straw Return Methods on the Soil Carbon Pool and CO₂ Emission in a Reclaimed Mine Spoil in Shanxi Province, China. *Soil Tillage Res.* **2019**, *195*, 104361. [\[CrossRef\]](#)
87. Sanger, A.; Geisseler, D.; Ludwig, B. Effects of Moisture and Temperature on Greenhouse Gas Emissions and C and N Leaching Losses in Soil Treated with Biogas Slurry. *Biol. Fertil. Soils* **2011**, *47*, 249–259. [\[CrossRef\]](#)
88. Rumpel, C.; Kogel-Knabner, I. Deep Soil Organic Matter—A Key but Poorly Understood Component of Terrestrial C Cycle. *Plant Soil* **2011**, *338*, 143–158. [\[CrossRef\]](#)
89. Levin, K.S.; Auerswald, K.; Reents, H.J.; Hulsbergen, K.-J. Effects of Organic Energy Crop Rotations and Fertilisation with the Liquid Digestate Phase on Organic Carbon in the Topsoil. *Agronomy* **2021**, *11*, 1393. [\[CrossRef\]](#)
90. Moller, K.; Stinner, W. Effects of Different Manuring Systems with and without Biogas Digestion on Soil Mineral Nitrogen Content and on Gaseous Nitrogen Losses (Ammonia, Nitrous Oxides). *Eur. J. Agron.* **2009**, *30*, 1–16. [\[CrossRef\]](#)
91. Alburquerque, J.A.; De La Fuente, C.; Bernal, M.P. Chemical Properties of Anaerobic Digestates Affecting C and N Dynamics in Amended Soils. *Agric. Ecosyst. Environ.* **2012**, *160*, 15–22. [\[CrossRef\]](#)
92. Alburquerque, J.A.; De La Fuente, C.; Campoy, M.; Carrasco, L.; Najera, I.; Baixauli, C.; Caravaca, F.; Roldan, A.; Cegarra, J.; Bernal, M.P. Agricultural Use of Digestate for Horticultural Crop Production and Improvement of Soil Properties. *Eur. J. Agron.* **2012**, *43*, 119–128. [\[CrossRef\]](#)
93. Makadi, M.; Tomocsik, A.; Orosz, V. Digestate: A New Nutrient Source—Review. In *Biogas*; Kumar, S., Ed.; IntechOpen: Rijeka, Croatia, 2012.
94. Levavasseur, F.; Le Roux, C.; Kouakou, P.; Jean-Baptiste, V.; Houot, S. High Nitrogen Availability but Limited Potential Carbon Storage in Anaerobic Digestates from Cover Crops. *J. Soil Sci. Plant Nutr.* **2022**, *22*, 2891–2896. [\[CrossRef\]](#)
95. Fouda, S.; Von Tucher, S.; Lichti, F.; Schmidhalter, U. Nitrogen Availability of Various Biogas Residues Applied to Ryegrass. *J. Plant Nutr. Soil Sci.* **2013**, *176*, 572–584. [\[CrossRef\]](#)
96. Follett, R.F. Chapter 2—Transformation and Transport Processes of Nitrogen in Agricultural Systems. In *Nitrogen in the Environment*, 2nd ed.; Hatfield, J.L., Follett, R.F., Eds.; Academic Press: San Diego, CA, USA, 2008; pp. 19–50. ISBN 978-0-12-374347-3.
97. Ren, A.-T.; Abbott, L.K.; Chen, Y.; Xiong, Y.-C.; Mickan, B.S. Nutrient Recovery from Anaerobic Digestion of Food Waste: Impacts of Digestate on Plant Growth and Rhizosphere Bacterial Community Composition and Potential Function in Ryegrass. *Biol. Fertil. Soils* **2020**, *56*, 973–989. [\[CrossRef\]](#)
98. Makadi, M.; Szegi, T.; Tomocsik, A.; Orosz, V.; Micheli, E.; Ferenczy, A.; Posta, K.; Biro, B. Impact of Digestate Application on Chemical and Microbiological Properties of Two Different Textured Soils. *Commun. Soil Sci. Plant Anal.* **2016**, *47*, 167–178. [\[CrossRef\]](#)
99. Gomez-Brandon, M.; Juarez, M.F.-D.; Zangerle, M.; Insam, H. Effects of Digestate on Soil Chemical and Microbiological Properties: A Comparative Study with Compost and Vermicompost. *J. Hazard. Mater.* **2016**, *302*, 267–274. [\[CrossRef\]](#) [\[PubMed\]](#)
100. Tsachidou, B.; Scheuren, M.; Gennen, J.; Debbaut, V.; Toussaint, B.; Hissler, C.; George, I.; Delfosse, P. Biogas Residues in Substitution for Chemical Fertilizers: A Comparative Study on a Grassland in the Walloon Region. *Sci. Total Environ.* **2019**, *666*, 212–225. [\[CrossRef\]](#)
101. Capra, F.; Abalos, D.; Maris, S.C.; Ardeni, F.; Lommi, M.; Tabaglio, V.; Fiorini, A. Towards Efficient N Cycling in Intensive Maize: Role of Cover Crops and Application Methods of Digestate Liquid Fraction. *GCB Bioenergy* **2023**, *15*, 867–885. [\[CrossRef\]](#)
102. Haraldsen, T.K.; Andersen, U.; Krogstad, T.; Sorheim, R. Liquid Digestate from Anaerobic Treatment of Source-Separated Household Waste as Fertilizer to Barley. *Waste Manag. Res. J. Sustain. Circ. Econ.* **2011**, *29*, 1271–1276. [\[CrossRef\]](#)
103. Sigurnjak, I.; Vaneekhaute, C.; Michels, E.; Ryckaert, B.; Ghekiere, G.; Tack, F.M.G.; Meers, E. Fertilizer Performance of Liquid Fraction of Digestate as Synthetic Nitrogen Substitute in Silage Maize Cultivation for Three Consecutive Years. *Sci. Total Environ.* **2017**, *599–600*, 1885–1894. [\[CrossRef\]](#)
104. Walsh, J.J.; Jones, D.L.; Edwards-Jones, G.; Williams, A.P. Replacing Inorganic Fertilizer with Anaerobic Digestate May Maintain Agricultural Productivity at Less Environmental Cost. *J. Plant Nutr. Soil Sci.* **2012**, *175*, 840–845. [\[CrossRef\]](#)

105. Zilio, M.; Pigoli, A.; Rizzi, B.; Goglio, A.; Tambone, F.; Giordano, A.; Maretto, L.; Squartini, A.; Stevanato, P.; Meers, E.; et al. Nitrogen Dynamics in Soils Fertilized with Digestate and Mineral Fertilizers: A Full Field Approach. *Sci. Total Environ.* **2023**, *868*, 161500. [\[CrossRef\]](#)
106. Tiong, Y.W.; Sharma, P.; Xu, S.; Bu, J.; An, S.; Foo, J.B.L.; Wee, B.K.; Wang, Y.; Lee, J.T.E.; Zhang, J.; et al. Enhancing Sustainable Crop Cultivation: The Impact of Renewable Soil Amendments and Digestate Fertilizer on Crop Growth and Nutrient Composition. *Environ. Pollut.* **2024**, *342*, 123132. [\[CrossRef\]](#)
107. Kovács, A.B.; Krempfer, R.; Kincses, I.; Szabó, A. Influences of Ammonium-Nitrate, Food Waste Compost and Bacterial Fertilizer on Soluble Soil Nitrogen Forms and on the Growth of Carrot (*Daucus carota* L.). *Eurasian J. Soil Sci.* **2014**, *3*, 95–100. [\[CrossRef\]](#)
108. Hupfauf, S.; Bachmann, S.; Fernández-Delgado Juárez, M.; Insam, H.; Eichler-Löbermann, B. Biogas Digestates Affect Crop P Uptake and Soil Microbial Community Composition. *Sci. Total Environ.* **2016**, *542*, 1144–1154. [\[CrossRef\]](#) [\[PubMed\]](#)
109. Katakai, S.; Hazarika, S.; Baruah, D.C. By-Products of Bioenergy Systems (Anaerobic Digestion and Gasification) as Sources of Plant Nutrients: Scope of Processed Application and Effect on Soil and Crop. *J. Mater. Cycles Waste Manag.* **2019**, *21*, 556–572. [\[CrossRef\]](#)
110. Valentinuzzi, F.; Cavani, L.; Porfido, C.; Terzano, R.; Pii, Y.; Cesco, S.; Marzadori, C.; Mimmo, T. The Fertilising Potential of Manure-Based Biogas Fermentation Residues: Pelleted vs. Liquid Digestate. *Heliyon* **2020**, *6*, e03325. [\[CrossRef\]](#)
111. Yu, F.-B.; Luo, X.-P.; Song, C.-F.; Zhang, M.-X.; Shan, S.-D. Concentrated Biogas Slurry Enhanced Soil Fertility and Tomato Quality. *Acta Agric. Scand. Sect. B-Plant Soil Sci.* **2010**, *60*, 262–268. [\[CrossRef\]](#)
112. Slepėtėnė, A.; Volungevičius, J.; Jurgutis, L.; Liaudanskienė, I.; Amalevičiūtė-Volunge, K.; Slepėtys, J.; Ceseviciene, J. The Potential of Digestate as a Biofertilizer in Eroded Soils of Lithuania. *Waste Manag.* **2020**, *102*, 441–451. [\[CrossRef\]](#)
113. Koszel, M.; Lorencowicz, E. Agricultural Use of Biogas Digestate as a Replacement Fertilizers. *Agric. Agric. Sci. Procedia* **2015**, *7*, 119–124. [\[CrossRef\]](#)
114. Kovács, A.B.; Kincses, I.; Krempfer, R.; Szoflikanics, A. Effects of Organic Fertilizers on the Growth and Nutrient Uptake of Cucumber (*Cucumis sativus* L.) Seedlings. *Ann. Acad. Romanian Sci. Ser. Agric. Silv. Vet. Med. Sci.* **2017**, *6*, 136–143.
115. Vanden Nest, T.; Ruysschaert, G.; Vandecasteele, B.; Cougnon, M.; Merckx, R.; Reheul, D. P Availability and P Leaching after Reducing the Mineral P Fertilization and the Use of Digestate Products as New Organic Fertilizers in a 4-Year Field Trial with High P Status. *Agric. Ecosyst. Environ.* **2015**, *202*, 56–67. [\[CrossRef\]](#)
116. Weimers, K.; Bergstrand, K.-J.; Hultberg, M.; Asp, H. Liquid Anaerobic Digestate as Sole Nutrient Source in Soilless Horticulture—Or Spiked with Mineral Nutrients for Improved Plant Growth. *Front. Plant Sci.* **2022**, *13*, 770179. [\[CrossRef\]](#)
117. Grigatti, M.; Boanini, E.; Cavani, L.; Ciavatta, C.; Marzadori, C. Phosphorus in Digestate-Based Compost: Chemical Speciation and Plant-Availability. *Waste Biomass Valorization* **2015**, *6*, 481–493. [\[CrossRef\]](#)
118. Menezes-Blackburn, D.; Paredes, C.; Zhang, H.; Giles, C.D.; Darch, T.; Stutter, M.; George, T.S.; Shand, C.; Lumsdon, D.; Cooper, P.; et al. Organic Acids Regulation of Chemical–Microbial Phosphorus Transformations in Soils. *Environ. Sci. Technol.* **2016**, *50*, 11521–11531. [\[CrossRef\]](#) [\[PubMed\]](#)
119. Singh, K.P.; Suman, A.; Singh, P.N.; Srivastava, T.K. Improving Quality of Sugarcane-Growing Soils by Organic Amendments under Subtropical Climatic Conditions of India. *Biol. Fertil. Soils* **2007**, *44*, 367–376. [\[CrossRef\]](#)
120. Bougnom, B.P.; Niederkofler, C.; Knapp, B.A.; Stimpfl, E.; Insam, H. Residues from Renewable Energy Production: Their Value for Fertilizing Pastures. *Biomass Bioenergy* **2012**, *39*, 290–295. [\[CrossRef\]](#)
121. Rolka, E.; Wyszowski, M.; Żołnowski, A.C.; Skorwider-Namiołko, A.; Szostek, R.; Wyzlic, K.; Borowski, M. Digestate from an Agricultural Biogas Plant as a Factor Shaping Soil Properties. *Agronomy* **2024**, *14*, 1528. [\[CrossRef\]](#)
122. Brtnický, M.; Kintl, A.; Holatko, J.; Hammerschmidt, T.; Mustafa, A.; Kucerik, J.; Vitez, T.; Prichystalova, J.; Baltazar, T.; Elbl, J. EFFECT of Digestates Derived from the Fermentation of Maize-Legume Intercropped Culture and Maize Monoculture Application on Soil Properties and Plant Biomass Production. *Chem. Biol. Technol. Agric.* **2022**, *9*, 43. [\[CrossRef\]](#)
123. Meng, X.; Zeng, B.; Wang, P.; Li, J.; Cui, R.; Ren, L. Food Waste Anaerobic Biogas Slurry as Fertilizer: Potential Salinization on Different Soil Layer and Effect on Rhizobacteria Community. *Waste Manag.* **2022**, *144*, 490–501. [\[CrossRef\]](#)
124. Jin, K.; Ran, Y.; Alengebawy, A.; Yang, G.; Jia, S.; Ai, P. Agro-Environmental Sustainability of Using Digestate Fertilizer for Solanaceous and Leafy Vegetables Cultivation: Insights on Fertilizer Efficiency and Risk Assessment. *J. Environ. Manag.* **2022**, *320*, 115895. [\[CrossRef\]](#)
125. Voelkner, A.; Holthusen, D.; Horn, R. Determination of Soil Dispersion Caused by Anaerobic Digestates: Interferences of pH and Soil Charge with Regard to Soil Texture and Water Content. *J. Soils Sediments* **2015**, *15*, 1491–1499. [\[CrossRef\]](#)
126. Odlare, M.; Pell, M.; Svensson, K. Changes in Soil Chemical and Microbiological Properties during 4 Years of Application of Various Organic Residues. *Waste Manag.* **2008**, *28*, 1246–1253. [\[CrossRef\]](#)
127. Pastorelli, R.; Valboa, G.; Lagomarsino, A.; Fabiani, A.; Simoncini, S.; Zaghi, M.; Vignozzi, N. Recycling Biogas Digestate from Energy Crops: Effects on Soil Properties and Crop Productivity. *Appl. Sci.* **2021**, *11*, 750. [\[CrossRef\]](#)
128. Lončarić, Z.; Vukobratović, M.; Ragalyi, P.; Filep, T.; Popovic, B.; Krunoslav, K.; Vukobratović, Ž. Computer Model for Organic Fertilizer Evaluation. *Poljopr. Osijek* **2009**, *15*, 38–46.

129. Shilpi, S.; Lamb, D.; Bolan, N.; Seshadri, B.; Choppala, G.; Naidu, R. Waste to Watt: Anaerobic Digestion of Wastewater Irrigated Biomass for Energy and Fertiliser Production. *J. Environ. Manag.* **2019**, *239*, 73–83. [\[CrossRef\]](#) [\[PubMed\]](#)
130. Voelkner, A.; Ohl, S.; Holthusen, D.; Hartung, E.; Dörner, J.; Horn, R. Impact of Mechanically Pre-Treated Anaerobic Digestates on Soil Properties. *J. Soil Sci. Plant Nutr.* **2015**, *15*, 882–895. [\[CrossRef\]](#)
131. Voelkner, A.; Holthusen, D.; Horn, R. Influence of Homogenized Residues of Anaerobic Digestate on the Physicochemical Properties of Differently Textured Soils. *J. Plant Nutr. Soil Sci.* **2015**, *178*, 261–269. [\[CrossRef\]](#)
132. Singh, B.; Chatterjee, A.; Chatterjee, R.; Chatterjee, M.; Banerjee, S. Molecular Mechanisms of Calcium Inducing Salt Tolerance in Rice: Ameliorative Interaction between CBL4 and P5CR Proteins. *Agrokém. És Talajt.* **2024**, *73*, 20–41. [\[CrossRef\]](#)
133. Jamison, J.; Khanal, S.K.; Nguyen, N.H.; Deenik, J.L. Assessing the Effects of Digestates and Combinations of Digestates and Fertilizer on Yield and Nutrient Use of Brassica Juncea (Kai Choy). *Agronomy* **2021**, *11*, 509. [\[CrossRef\]](#)
134. Tampio, E.; Salo, T.; Rintala, J. Agronomic Characteristics of Five Different Urban Waste Digestates. *J. Environ. Manag.* **2016**, *169*, 293–302. [\[CrossRef\]](#)
135. Mortola, N.; Romaniuk, R.; Cosentino, V.; Eiza, M.; Carfagno, P.; Rizzo, P.; Bres, P.; Riera, N.; Roba, M.; Butti, M.; et al. Potential Use of a Poultry Manure Digestate as a Biofertiliser: Evaluation of Soil Properties and Lactuca Sativa Growth. *Pedosphere* **2019**, *29*, 60–69. [\[CrossRef\]](#)
136. Ferrarezi, R.S.; Lin, X.; Gonzalez Neira, A.C.; Tabay Zambon, F.; Hu, H.; Wang, X.; Huang, J.-H.; Fan, G. Substrate pH Influences the Nutrient Absorption and Rhizosphere Microbiome of Huanglongbing-Affected Grapefruit Plants. *Front. Plant Sci.* **2022**, *13*, 856937. [\[CrossRef\]](#)
137. Simon, F.W.; de Castilhos Junior, A.B.; Loss, A.; Malinowski, C.; Matias, M.S. Effects of Food Waste Digested Materials on Lactuca Sativa Growth and Soil Composition. *Int. J. Environ. Sci. Technol.* **2023**, *20*, 9013–9028. [\[CrossRef\]](#)
138. Liu, W.; Yao, B.; Xu, Y.; Dai, S.; Wang, M.; Ma, J.; Ye, Z.; Liu, D. Biogas Digestate as a Potential Nitrogen Source Enhances Soil Fertility, Rice Nitrogen Metabolism and Yield. *Field Crops Res.* **2024**, *318*, 109568. [\[CrossRef\]](#)
139. Bloom, P.R.; Skyllberg, U. Soil pH and pH Buffering. In *Handbook of Soil Sciences: Properties and Processes*; CRC Press: Boca Raton, FL, USA, 2011; pp. 19–1–19–14. ISBN 978-0-429-09598-6.
140. Garg, R.N.; Pathak, H.; Das, D.K.; Tomar, R.K. Use of Flyash and Biogas Slurry for Improving Wheat Yield and Physical Properties of Soil. *Environ. Monit. Assess.* **2005**, *107*, 1–9. [\[CrossRef\]](#) [\[PubMed\]](#)
141. El-Bakhshawan, M.K.; Minyawi, M.; Aboukarima, A.M. Impact of adding biogas liquid slurry on soil surface on penetration resistance. *Misr J. Agric. Eng.* **2017**, *34*, 525–540. [\[CrossRef\]](#)
142. Jaša, S.; Badalíková, B.; Červinka, J. Influence of Digestate on Physical Properties of Soil in ZD Budišov. *Acta Univ. Agric. Silv. Mendel. Brun.* **2019**, *67*, 75–83. [\[CrossRef\]](#)
143. Malav, L.C.; Khan, S.; Gupta, N.; Kumar, S.; Bhattacharyya, R.; Malav, M.K. Effect of Biogas Slurry and Urea on Soil Health. *J. Agric. Phys.* **2015**, *15*, 55–62.
144. Mayerová, M.; Šimon, T.; Stehlík, M.; Madaras, M.; Koubová, M.; Smatanová, M. Long-Term Application of Biogas Digestate Improves Soil Physical Properties. *Soil Tillage Res.* **2023**, *231*, 105715. [\[CrossRef\]](#)
145. Zheng, X.; Fan, J.; Xu, L.; Zhou, J. Effects of Combined Application of Biogas Slurry and Chemical Fertilizer on Soil Aggregation and C/N Distribution in an Ultisol. *PLoS ONE* **2017**, *12*, e0170491. [\[CrossRef\]](#)
146. Tang, J.; Yin, J.; Davy, A.J.; Pan, F.; Han, X.; Huang, S.; Wu, D. Biogas Slurry as an Alternative to Chemical Fertilizer: Changes in Soil Properties and Microbial Communities of Fluvo-Aquic Soil in the North China Plain. *Sustainability* **2022**, *14*, 15099. [\[CrossRef\]](#)
147. Dai, H.; Chen, Y.; Liu, K.; Li, Z.; Qian, X.; Zang, H.; Yang, X.; Zhao, Y.; Shen, Y.; Li, Z.; et al. Water-Stable Aggregates and Carbon Accumulation in Barren Sandy Soil Depend on Organic Amendment Method: A Three-Year Field Study. *J. Clean. Prod.* **2019**, *212*, 393–400. [\[CrossRef\]](#)
148. Koch, S.; Liu, H.; Lenz, C.; Eichler-Löbermann, B.; Vogel, H.-J.; Lennartz, B. Soil Structure and Solute Transport Pathways in Biogas Digestate-Amended Soils. *Soil Tillage Res.* **2024**, *240*, 106074. [\[CrossRef\]](#)
149. Palupi, C.C.; Ngadisih, N.; Karyadi, J.N.W.; Tirtalistyani, R.; Heikal Ismail, M.; Mawandha, H.G. Effect of Biogas Slurry Fertilizer on Dynamics of Soil Consistency and Tillage Power Requirement. *J. Ilm. Rekayasa Pertan. Dan Biosist.* **2022**, *10*, 14–23. [\[CrossRef\]](#)
150. Musse, Z.A.; Yoseph Samago, T.; Beshir, H.M. Effect of Liquid Bio-Slurry and Nitrogen Rates on Soil Physico-Chemical Properties and Quality of Green Bean (*Phaseolus vulgaris* L.) at Hawassa Southern Ethiopia. *J. Plant Interact.* **2020**, *15*, 207–212. [\[CrossRef\]](#)
151. Angst, G.; Mueller, K.E.; Nierop, K.G.J.; Simpson, M.J. Plant- or Microbial-Derived? A Review on the Molecular Composition of Stabilized Soil Organic Matter. *Soil Biol. Biochem.* **2021**, *156*, 108189. [\[CrossRef\]](#)
152. Wang, B.; An, S.; Liang, C.; Liu, Y.; Kuzyakov, Y. Microbial Necromass as the Source of Soil Organic Carbon in Global Ecosystems. *Soil Biol. Biochem.* **2021**, *162*, 108422. [\[CrossRef\]](#)
153. Chen, Z.; Ma, J.; Ma, J.; Ye, J.; Yu, Q.; Zou, P.; Sun, W.; Lin, H.; Wang, F.; Zhao, X.; et al. Long-Term Biogas Slurry Application Increases Microbial Necromass but Not Plant Lignin Contribution to Soil Organic Carbon in Paddy Soils as Regulated by Fungal Community. *Waste Manag.* **2024**, *175*, 254–264. [\[CrossRef\]](#)

154. Ernst, G.; Müller, A.; Göhler, H.; Emmerling, C. C and N Turnover of Fermented Residues from Biogas Plants in Soil in the Presence of Three Different Earthworm Species (*Lumbricus Terrestris*, *Aporrectodea Longa*, *Aporrectodea Caliginosa*). *Soil Biol. Biochem.* **2008**, *40*, 1413–1420. [\[CrossRef\]](#)
155. Thomsen, I.K.; Olesen, J.E.; Møller, H.B.; Sørensen, P.; Christensen, B.T. Carbon Dynamics and Retention in Soil after Anaerobic Digestion of Dairy Cattle Feed and Faeces. *Soil Biol. Biochem.* **2013**, *58*, 82–87. [\[CrossRef\]](#)
156. Ahlberg-Eliasson, K.; Liu, T.; Nadeau, E.; Schnürer, A. Forage Types and Origin of Manure in Codigestion Affect Methane Yield and Microbial Community Structure. *Grass Forage Sci.* **2018**, *73*, 740–757. [\[CrossRef\]](#)
157. Wentzel, S.; Joergensen, R.G. Quantitative Microbial Indices in Biogas and Raw Cattle Slurries. *Eng. Life Sci.* **2016**, *16*, 231–237. [\[CrossRef\]](#)
158. Kolář, L.; Kužel, S.; Peterka, J.; Štindl, P.; Plát, V. Agrochemical Value of Organic Matter of Fermenter Wastes in Biogas Production. *Plant Soil Environ.* **2008**, *54*, 321–328. [\[CrossRef\]](#)
159. Coban, H.; Miltner, A.; Elling, F.J.; Hinrichs, K.-U.; Kästner, M. The Contribution of Biogas Residues to Soil Organic Matter Formation and CO₂ Emissions in an Arable Soil. *Soil Biol. Biochem.* **2015**, *86*, 108–115. [\[CrossRef\]](#)
160. Insam, H.; Gómez-Brandón, M.; Ascher, J. Manure-Based Biogas Fermentation Residues—Friend or Foe of Soil Fertility? *Soil Biol. Biochem.* **2015**, *84*, 1–14. [\[CrossRef\]](#)
161. Stumpe, B.; Werner, S.; Jung, R.; Heinze, S.; Juschke, E.; Strippel, C.; Marschner, B. Organic Carbon Dynamics and Enzyme Activities in Agricultural Soils Amended with Biogas Slurry, Liquid Manure and Sewage Sludge. *Agric. Sci.* **2012**, *3*, 104–113. [\[CrossRef\]](#)
162. Sapp, M.; Harrison, M.; Hany, U.; Charlton, A.; Thwaites, R. Comparing the Effect of Digestate and Chemical Fertiliser on Soil Bacteria. *Appl. Soil Ecol.* **2015**, *86*, 1–9. [\[CrossRef\]](#)
163. Jothi, G.; Pugalendhi, S.; Poornima, K.; Rajendran, G. Management of Root-Knot Nematode in Tomato *Lycopersicon Esculentum*, Mill., with Biogas Slurry. *Bioresour. Technol.* **2003**, *89*, 169–170. [\[CrossRef\]](#)
164. Westphal, A.; Kücke, M.; Heuer, H. Soil Amendment with Digestate from Bio-Energy Fermenters for Mitigating Damage to Beta Vulgaris Subsp. by *Heterodera Schachtii*. *Appl. Soil Ecol.* **2016**, *99*, 129–136. [\[CrossRef\]](#)
165. Moinard, V.; Redondi, C.; Etiévant, V.; Savoie, A.; Duchene, D.; Pelosi, C.; Houot, S.; Capowiez, Y. Short- and Long-Term Impacts of Anaerobic Digestate Spreading on Earthworms in Cropped Soils. *Appl. Soil Ecol.* **2021**, *168*, 104149. [\[CrossRef\]](#)
166. Garbini, G.L.; Grenni, P.; Rauseo, J.; Patrolecco, L.; Pescatore, T.; Spataro, F.; Barra Caracciolo, A. Insights into Structure and Functioning of a Soil Microbial Community Amended with Cattle Manure Digestate and Sulfamethoxazole. *J. Soils Sediments* **2022**, *22*, 2158–2173. [\[CrossRef\]](#)
167. Garcia-Sánchez, M.; Garcia-Romera, I.; Cajthaml, T.; Tlustoš, P.; Száková, J. Changes in Soil Microbial Community Functionality and Structure in a Metal-Polluted Site: The Effect of Digestate and Fly Ash Applications. *J. Environ. Manag.* **2015**, *162*, 63–73. [\[CrossRef\]](#)
168. Meng, X.; Ma, C.; Petersen, S.O. Sensitive Control of N₂O Emissions and Microbial Community Dynamics by Organic Fertilizer and Soil Interactions. *Biol. Fertil. Soils* **2022**, *58*, 771–788. [\[CrossRef\]](#)
169. Cattin, M.; Semple, K.T.; Stutter, M.; Romano, G.; Lag-Brotons, A.J.; Parry, C.; Surridge, B.W.J. Changes in Microbial Utilization and Fate of Soil Carbon Following the Addition of Different Fractions of Anaerobic Digestate to Soils. *Eur. J. Soil Sci.* **2021**, *72*, 2398–2413. [\[CrossRef\]](#)
170. Baldasso, V.; Tomasino, M.P.; Sayen, S.; Guillon, E.; Frunzo, L.; Gomes, C.A.R.; Alves, M.J.; Castro, R.; Mucha, A.P.; Almeida, C.M.R. Effects of Digestate Soil Amendment on the Fate of Trace Metals and on the Soil Microbial Community. *Environ. Pollut.* **2025**, *371*, 125961. [\[CrossRef\]](#) [\[PubMed\]](#)
171. Madegwa, Y.M.; Uchida, Y. Liming Improves the Stability of Soil Microbial Community Structures against the Application of Digestate Made from Dairy Wastes. *J. Environ. Manag.* **2021**, *297*, 113356. [\[CrossRef\]](#) [\[PubMed\]](#)
172. Mora-Salguero, D.; Montenach, D.; Gilles, M.; Jean-Baptiste, V.; Sadet-Bourgeteau, S. Long-Term Effects of Combining Anaerobic Digestate with Other Organic Waste Products on Soil Microbial Communities. *Front. Microbiol.* **2025**, *15*, 1490034. [\[CrossRef\]](#)
173. Nikolaidou, C.; Mola, M.; Papakostas, S.; Aschonitis, V.G.; Monokrousos, N.; Kougias, P.G. The Effect of Anaerobic Digestate as an Organic Soil Fertilizer on the Diversity and Structure of the Indigenous Soil Microbial and Nematode Communities. *Environ. Sci. Pollut. Res.* **2024**. [\[CrossRef\]](#)
174. Pezzolla, D.; Marconi, G.; Turchetti, B.; Zadra, C.; Agnelli, A.; Veronesi, F.; Onofri, A.; Benucci, G.M.N.; Buzzini, P.; Albertini, E.; et al. Influence of Exogenous Organic Matter on Prokaryotic and Eukaryotic Microbiota in an Agricultural Soil. A Multidisciplinary Approach. *Soil Biol. Biochem.* **2015**, *82*, 9–20. [\[CrossRef\]](#)
175. Barduca, L.; Wentzel, S.; Schmidt, R.; Malagoli, M.; Joergensen, R.G. Mineralisation of Distinct Biogas Digestate Qualities Directly after Application to Soil. *Biol. Fertil. Soils* **2021**, *57*, 235–243. [\[CrossRef\]](#)
176. Yu, X.-Y.; Zhu, Y.-J.; Jin, L.; Wang, B.-T.; Xu, X.; Zou, X.; Ruan, H.-H.; Jin, F.-J. Contrasting Responses of Fungal and Bacterial Communities to Biogas Slurry Addition in Rhizospheric Soil of Poplar Plantations. *Appl. Soil Ecol.* **2022**, *175*, 104427. [\[CrossRef\]](#)

177. Xu, M.; Xian, Y.; Wu, J.; Gu, Y.; Yang, G.; Zhang, X.; Peng, H.; Yu, X.; Xiao, Y.; Li, L. Effect of Biogas Slurry Addition on Soil Properties, Yields, and Bacterial Composition in the Rice-Rape Rotation Ecosystem over 3 Years. *J. Soils Sediments* **2019**, *19*, 2534–2542. [\[CrossRef\]](#)
178. Chen, Z.; Wang, Q.; Ma, J.; Chapman, S.; Zou, P.; Ye, J.; Yu, Q.; Sun, W.; Lin, H.; Jiang, L. Soil Microbial Activity and Community Composition as Influenced by Application of Pig Biogas Slurry in Paddy Field in Southeast China. *Paddy Water Environ.* **2020**, *18*, 15–25. [\[CrossRef\]](#)
179. Wang, C.; Liu, J.; Wang, C.; Zhao, X.; Wu, K.; Yang, B.; Yin, F.; Zhang, W. Biogas Slurry Application Alters Soil Properties, Reshapes the Soil Microbial Community, and Alleviates Root Rot of *Panax Notoginseng*. *PeerJ* **2022**, *10*, e13770. [\[CrossRef\]](#) [\[PubMed\]](#)
180. Czatzkowska, M.; Rolbiecki, D.; Korzeniewska, E.; Harnisz, M. Heavy Metal and Antimicrobial Residue Levels in Various Types of Digestate from Biogas Plants—A Review. *Sustainability* **2025**, *17*, 416. [\[CrossRef\]](#)
181. Thanh, P.M.; Ketheesan, B.; Yan, Z.; Stuckey, D. Trace Metal Speciation and Bioavailability in Anaerobic Digestion: A Review. *Biotechnol. Adv.* **2016**, *34*, 122–136. [\[CrossRef\]](#) [\[PubMed\]](#)
182. Tang, Y.; Wang, L.; Carswell, A.; Misselbrook, T.; Shen, J.; Han, J. Fate and Transfer of Heavy Metals Following Repeated Biogas Slurry Application in a Rice-Wheat Crop Rotation. *J. Environ. Manag.* **2020**, *270*, 110938. [\[CrossRef\]](#)
183. Derehajtó, S.; Tymińska, M.; Skibko, Z.; Borusiewicz, A.; Romaniuk, W.; Kuboń, M.; Olech, E.; Koszel, M. Heavy Metal Content in Substrates in Agricultural Biogas Plants. *Agric. Eng.* **2023**, *27*, 315–329. [\[CrossRef\]](#)
184. Bian, B.; Wu, H.S.; Zhou, L.J. Contamination and Risk Assessment of Heavy Metals in Soils Irrigated with Biogas Slurry: A Case Study of Taihu Basin. *Environ. Monit. Assess.* **2015**, *187*, 155. [\[CrossRef\]](#)
185. Lu, J.; Jiang, L.; Chen, D.; Toyota, K.; Strong, P.J.; Wang, H.; Hirasawa, T. Decontamination of Anaerobically Digested Slurry in a Paddy Field Ecosystem in Jiaying Region of China. *Agric. Ecosyst. Environ.* **2012**, *146*, 13–22. [\[CrossRef\]](#)
186. Wang, Q.; Chen, Z.; Zhao, J.; Ma, J.; Yu, Q.; Zou, P.; Lin, H.; Ma, J. Fate of Heavy Metals and Bacterial Community Composition Following Biogas Slurry Application in a Single Rice Cropping System. *J. Soils Sediments* **2022**, *22*, 968–981. [\[CrossRef\]](#)
187. Wang, Q.; Huang, Q.; Wang, J.; Li, H.; Qin, J.; Li, X.; Gouda, S.G.; Liu, Y.; Liu, Q.; Guo, G.; et al. Ecological Circular Agriculture: A Case Study Evaluating Biogas Slurry Applied to Rice in Two Soils. *Chemosphere* **2022**, *301*, 134628. [\[CrossRef\]](#)
188. Fang, W.; Wei, Y.; Liu, J. Comparative Characterization of Sewage Sludge Compost and Soil: Heavy Metal Leaching Characteristics. *J. Hazard. Mater.* **2016**, *310*, 1–10. [\[CrossRef\]](#)
189. Sánchez-Martín, M.J.; García-Delgado, M.; Lorenzo, L.F.; Rodríguez-Cruz, M.S.; Arienzo, M. Heavy Metals in Sewage Sludge Amended Soils Determined by Sequential Extractions as a Function of Incubation Time of Soils. *Geoderma* **2007**, *142*, 262–273. [\[CrossRef\]](#)
190. Pasquini, L.; Munoz, J.-F.; Pons, M.-N.; Yvon, J.; Dauchy, X.; France, X.; Le, N.D.; France-Lanord, C.; Görner, T. Occurrence of Eight Household Micropollutants in Urban Wastewater and Their Fate in a Wastewater Treatment Plant. Statistical Evaluation. *Sci. Total Environ.* **2014**, *481*, 459–468. [\[CrossRef\]](#) [\[PubMed\]](#)
191. Vodyanitskii, Y.N.; Yakovlev, A.S. Contamination of Soils and Groundwater with New Organic Micropollutants: A Review. *Eurasian Soil Sci.* **2016**, *49*, 560–569. [\[CrossRef\]](#)
192. García-Santiago, X.; Franco-Uría, A.; Omil, F.; Lema, J.M. Risk Assessment of Persistent Pharmaceuticals in Biosolids: Dealing with Uncertainty. *J. Hazard. Mater.* **2016**, *302*, 72–81. [\[CrossRef\]](#)
193. Chenxi, W.; Spongberg, A.L.; Witter, J.D. Determination of the Persistence of Pharmaceuticals in Biosolids Using Liquid-Chromatography Tandem Mass Spectrometry. *Chemosphere* **2008**, *73*, 511–518. [\[CrossRef\]](#)
194. Carvalho, P.N.; Basto, M.C.P.; Almeida, C.M.R.; Brix, H. A Review of Plant–Pharmaceutical Interactions: From Uptake and Effects in Crop Plants to Phytoremediation in Constructed Wetlands. *Environ. Sci. Pollut. Res.* **2014**, *21*, 11729–11763. [\[CrossRef\]](#)
195. Clarke, B.O.; Smith, S.R. Review of ‘Emerging’ Organic Contaminants in Biosolids and Assessment of International Research Priorities for the Agricultural Use of Biosolids. *Environ. Int.* **2011**, *37*, 226–247. [\[CrossRef\]](#)
196. García-Valcárcel, A.I.; Tadeo, J.L. Influence of Moisture on the Availability and Persistence of Clotrimazole and Fluconazole in Sludge-Amended Soil. *Environ. Toxicol. Chem.* **2012**, *31*, 501–507. [\[CrossRef\]](#)
197. Martín, J.; Camacho-Muñoz, M.D.; Santos, J.L.; Aparicio, I.; Alonso, E. Distribution and Temporal Evolution of Pharmaceutically Active Compounds alongside Sewage Sludge Treatment. Risk Assessment of Sludge Application onto Soils. *J. Environ. Manag.* **2012**, *102*, 18–25. [\[CrossRef\]](#)
198. Thiele-Bruhn, S. Pharmaceutical Antibiotic Compounds in Soils—A Review. *J. Plant Nutr. Soil Sci.* **2003**, *166*, 145–167. [\[CrossRef\]](#)
199. León, C.; Henríquez, C.; López, N.; Sanchez, G.; Pastén, B.; Baeza, P.; Ojeda, J. Inhibitory Effect of the Ascorbic Acid on Photodegradation of Pharmaceuticals Compounds Exposed to UV-B Radiation. *J. Photochem. Photobiol.* **2021**, *7*, 100035. [\[CrossRef\]](#)
200. Chen, W.-L.; Lin, S.-C.; Huang, C.-H.; Peng, S.-Y.; Ling, Y.S. Wide-Scope Screening for Pharmaceutically Active Substances in a Leafy Vegetable Cultivated under Biogas Slurry Irrigation. *Sci. Total Environ.* **2021**, *750*, 141519. [\[CrossRef\]](#) [\[PubMed\]](#)
201. Widyasari-Mehta, A.; Hartung, S.; Kreuzig, R. From the Application of Antibiotics to Antibiotic Residues in Liquid Manures and Digestates: A Screening Study in One European Center of Conventional Pig Husbandry. *J. Environ. Manag.* **2016**, *177*, 129–137. [\[CrossRef\]](#) [\[PubMed\]](#)

202. Lehmann, L.; Bloem, E. Antibiotic Residues in Substrates and Output Materials from Biogas Plants—Implications for Agriculture. *Chemosphere* **2021**, *278*, 130425. [\[CrossRef\]](#)
203. Ali, A.M.; Nesse, A.S.; Eich-Greatorex, S.; Sogn, T.A.; Aanrud, S.G.; Bunæs, J.A.A.; Lyche, J.L.; Kallenborn, R. Organic Contaminants of Emerging Concern in Norwegian Digestates from Biogas Production. *Environ. Sci. Process. Impacts* **2019**, *21*, 1498–1508. [\[CrossRef\]](#)
204. Rodriguez-Navas, C.; Björklund, E.; Halling-Sørensen, B.; Hansen, M. Biogas Final Digestive Byproduct Applied to Croplands as Fertilizer Contains High Levels of Steroid Hormones. *Environ. Pollut.* **2013**, *180*, 368–371. [\[CrossRef\]](#)
205. Nguyen, M.K.; Hadi, M.; Lin, C.; Nguyen, H.-L.; Thai, V.-B.; Hoang, H.-G.; Vo, D.-V.N.; Tran, H.-T. Microplastics in Sewage Sludge: Distribution, Toxicity, Identification Methods, and Engineered Technologies. *Chemosphere* **2022**, *308*, 136455. [\[CrossRef\]](#)
206. He, D.; Luo, Y.; Lu, S.; Liu, M.; Song, Y.; Lei, L. Microplastics in Soils: Analytical Methods, Pollution Characteristics and Ecological Risks. *TrAC Trends Anal. Chem.* **2018**, *109*, 163–172. [\[CrossRef\]](#)
207. Sutkar, P.R.; Gadewar, R.D.; Dhulap, V.P. Recent Trends in Degradation of Microplastics in the Environment: A State-of-the-Art Review. *J. Hazard. Mater. Adv.* **2023**, *11*, 100343. [\[CrossRef\]](#)
208. Ashrafi, A.; Liza, A.A.; Islam, M.N.; Billah, M.M.; Arafat, S.T.; Rahman, M.M.; Rahman, S.M. Microplastics Pollution: A Brief Review of Its Source and Abundance in Different Aquatic Ecosystems. *J. Hazard. Mater. Adv.* **2023**, *9*, 100215. [\[CrossRef\]](#)
209. Nanthini Devi, K.; Raju, P.; Santhanam, P.; Perumal, P. Impacts of Microplastics on Marine Organisms: Present Perspectives and the Way Forward. *Egypt. J. Aquat. Res.* **2022**, *48*, 205–209. [\[CrossRef\]](#)
210. Wang, X.; Zhang, Y.; Zhao, Y.; Zhang, L.; Zhang, X. Inhibition of Aged Microplastics and Leachates on Methane Production from Anaerobic Digestion of Sludge and Identification of Key Components. *J. Hazard. Mater.* **2023**, *446*, 130717. [\[CrossRef\]](#) [\[PubMed\]](#)
211. Sun, M.; Xiao, K.; Zhu, Y.; Ou, B.; Yu, W.; Liang, S.; Hou, H.; Yuan, S.; Gan, F.; Mi, R.; et al. Deciphering the Role of Microplastic Size on Anaerobic Sludge Digestion: Changes of Dissolved Organic Matter, Leaching Compounds and Microbial Community. *Environ. Res.* **2022**, *214*, 114032. [\[CrossRef\]](#) [\[PubMed\]](#)
212. Alimohammadi, M.; Demirer, G.N. Microplastics in Anaerobic Digestion: Occurrence, Impact, and Mitigation Strategies. *J. Environ. Health Sci. Eng.* **2024**, *22*, 397–411. [\[CrossRef\]](#)
213. Mahon, A.M.; O’Connell, B.; Healy, M.G.; O’Connor, I.; Officer, R.; Nash, R.; Morrison, L. Microplastics in Sewage Sludge: Effects of Treatment. *Environ. Sci. Technol.* **2017**, *51*, 810–818. [\[CrossRef\]](#)
214. Chand, R.; Rasmussen, L.A.; Tumlin, S.; Vollertsen, J. The Occurrence and Fate of Microplastics in a Mesophilic Anaerobic Digester Receiving Sewage Sludge, Grease, and Fatty Slurries. *Sci. Total Environ.* **2021**, *798*, 149287. [\[CrossRef\]](#)
215. Alvim, C.B.; Ferrer-Polonio, E.; Bes-Piá, M.A.; Mendoza-Roca, J.A.; Fernández-Navarro, J.; Alonso-Molina, J.L.; Amorós-Muñoz, I. Effect of Polystyrene Nanoplastics on the Activated Sludge Process Performance and Biomass Characteristics. A Laboratory Study with a Sequencing Batch Reactor. *J. Environ. Manag.* **2023**, *329*, 117131. [\[CrossRef\]](#)
216. Raju, S.; Carbery, M.; Kuttykattil, A.; Senathirajah, K.; Subashchandrabose, S.R.; Evans, G.; Thavamani, P. Transport and Fate of Microplastics in Wastewater Treatment Plants: Implications to Environmental Health. *Rev. Environ. Sci. Biotechnol.* **2018**, *17*, 637–653. [\[CrossRef\]](#)
217. Liu, L.; Xu, M.; Ye, Y.; Zhang, B. On the Degradation of (Micro)Plastics: Degradation Methods, Influencing Factors, Environmental Impacts. *Sci. Total Environ.* **2022**, *806*, 151312. [\[CrossRef\]](#)
218. Chamas, A.; Moon, H.; Zheng, J.; Qiu, Y.; Tabassum, T.; Jang, J.H.; Abu-Omar, M.; Scott, S.L.; Suh, S. Degradation Rates of Plastics in the Environment. *ACS Sustain. Chem. Eng.* **2020**, *8*, 3494–3511. [\[CrossRef\]](#)
219. Aitken, M.D.; Sobsey, M.D.; Van Abel, N.A.; Blauth, K.E.; Singleton, D.R.; Crunk, P.L.; Nichols, C.; Walters, G.W.; Schneider, M. Inactivation of Escherichia Coli O157:H7 during Thermophilic Anaerobic Digestion of Manure from Dairy Cattle. *Water Res.* **2007**, *41*, 1659–1666. [\[CrossRef\]](#) [\[PubMed\]](#)
220. Sidhu, J.; Toze, S. Human Pathogens and Their Indicators in Biosolids: A Literature Review. *Environ. Int.* **2009**, *35*, 187–201. [\[CrossRef\]](#) [\[PubMed\]](#)
221. Goberna, M.; Podmirseg, S.M.; Waldhuber, S.; Knapp, B.A.; García, C.; Insam, H. Pathogenic Bacteria and Mineral N in Soils Following the Land Spreading of Biogas Digestates and Fresh Manure. *Appl. Soil Ecol.* **2011**, *49*, 18–25. [\[CrossRef\]](#)
222. Bonetta, S.; Ferretti, E.; Bonetta, S.; Fezia, G.; Carraro, E. Microbiological Contamination of Digested Products from Anaerobic Co-Digestion of Bovine Manure and Agricultural by-Products: Microbial Contamination of Digestate. *Lett. Appl. Microbiol.* **2011**, *53*, 552–557. [\[CrossRef\]](#)
223. Islam, M.; Biswas, P.; Sabuj, A.; Haque, Z.; Saha, C.; Alam, M.; Rahman, M.; Saha, S. Microbial Load in Bio-Slurry from Different Biogas Plants in Bangladesh. *J. Adv. Vet. Anim. Res.* **2019**, *6*, 376. [\[CrossRef\]](#)
224. Nag, R.; Whyte, P.; Markey, B.K.; O’Flaherty, V.; Bolton, D.; Fenton, O.; Richards, K.G.; Cummins, E. Ranking Hazards Pertaining to Human Health Concerns from Land Application of Anaerobic Digestate. *Sci. Total Environ.* **2020**, *710*, 136297. [\[CrossRef\]](#)
225. Massé, D.; Gilbert, Y.; Topp, E. Pathogen Removal in Farm-Scale Psychrophilic Anaerobic Digesters Processing Swine Manure. *Bioresour. Technol.* **2011**, *102*, 641–646. [\[CrossRef\]](#)

-
226. Pu, C.; Liu, H.; Ding, G.; Sun, Y.; Yu, X.; Chen, J.; Ren, J.; Gong, X. Impact of Direct Application of Biogas Slurry and Residue in Fields: In Situ Analysis of Antibiotic Resistance Genes from Pig Manure to Fields. *J. Hazard. Mater.* **2018**, *344*, 441–449. [[CrossRef](#)]
227. Lu, Y.; Li, J.; Meng, J.; Zhang, J.; Zhuang, H.; Zheng, G.; Xie, W.; Ping, L.; Shan, S. Long-Term Biogas Slurry Application Increased Antibiotics Accumulation and Antibiotic Resistance Genes (ARGs) Spread in Agricultural Soils with Different Properties. *Sci. Total Environ.* **2021**, *759*, 143473. [[CrossRef](#)]

Disclaimer/Publisher’s Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.